
EMCH 574 – Vibration and Waves Homework 01

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Due date TBD

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Assignment

- G signifies students that take this class for Graduate Credit; they must do all the items.
- UG students are not required to do the G work, but they may do it for bonus points.
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

Notes

- Show your work to obtain partial credit if appropriate.
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement [reference](#).
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga, and so on) [reference](#).
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit.
- Normalized plots are plots with the *y-axis* scaled between -1 and $+1$. A plotted variable is normalized by dividing it by its maximum absolute value.

Notations and Abbreviations

- BVP = boundary value problem.
- EOM = equation of motion.
- FBD = free body diagram.
- FRF = frequency response function.
- IVP = initial value problem.
- LHS = left hand side.
- NLM = Newton's law of motion.
- ODE = ordinary differential equation.
- PDE = partial differential equation.
- RHS = right hand side.

1 Pendulum

A.1 Analyze Pendulum

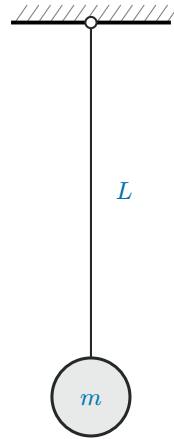


Figure 1: Pendulum of length L and mass m .

Consider a pendulum of length L and mass m as shown in Figure 1.

Do the following.

- (1) Draw the pendulum displaced by an angle θ from the vertical.
- (2) Draw free-body diagram (FBD).
- (3) Write Newton's law of motion (NLM) equation.
- (4) Apply small-angle approximation and deduce the equation of motion (EOM). Clearly state your assumptions.
- (5) Deduce the governing equation.
- (6) Assume $\theta(t) = e^{st}$ and substitute into the governing equation.
- (7) Deduce and solve the characteristic equation.
- (8) What type are the roots $s_{1,2}$? Choose one of the following options:
 - (a) Real.
 - (b) Imaginary.
 - (c) Complex.
- (9) Define the natural frequency ω_n .
- (10) Express the solutions s_1 and s_2 of the characteristic equation in terms of ω_n .
- (11) Write the solution in terms of exponential functions of $\omega_n t$.
- (12) Write the solution in terms of trigonometric functions of $\omega_n t$.
- (13) Write the Hz frequency f_n .

- (14) Write the period of oscillation τ .
- (15) Write the particle motion $u(t)$ in terms of:
- (a) Exponential functions.
 - (b) Trigonometric functions.
- (16) Solve the initial value problem (IVP) and find the response $u(t)$ for:
- (i) Initial displacement u_0 .
 - (ii) Initial velocity $\dot{u}_0$.
 - (iii) Simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$.

Worked solution

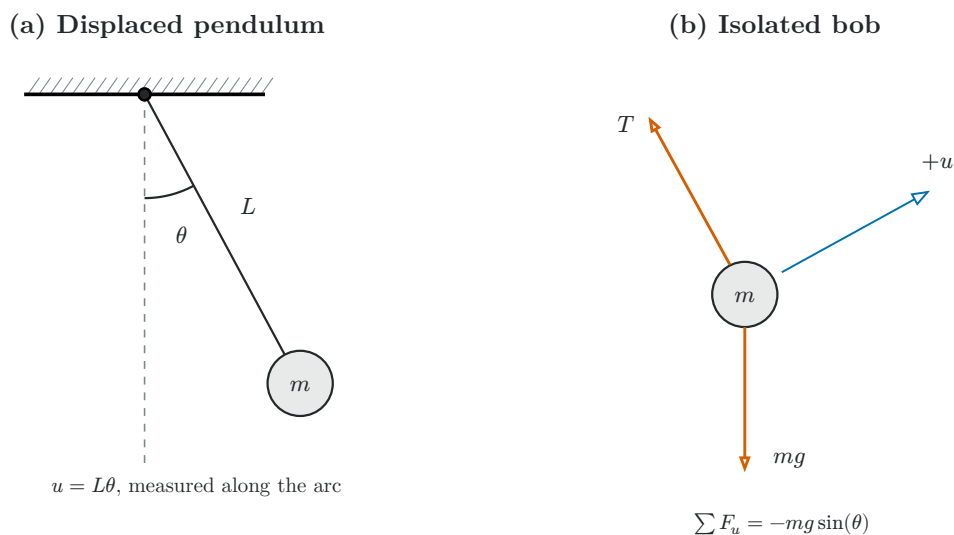


Figure 2: Free-body diagram for Problem A.1.

Assume a point mass, a massless inextensible rod or taut string of length L , a fixed frictionless pivot, uniform gravity, planar motion, and negligible air resistance. Measure θ from the downward vertical. Let $u = L\theta$ be arc displacement. This relation is exact for arc displacement; horizontal displacement is $L \sin \theta \approx u$ only for small angles.

The free-body diagram has weight mg downward and tension T toward the pivot. Tension has no tangential component. Gravity contributes the restoring tangential force $-mg \sin \theta$. Newton's law of motion therefore gives

$$mL\ddot{\theta} = -mg \sin \theta.$$

For $|\theta| \ll 1$ in radians, $\sin \theta \approx \theta$, so the linear equation of motion and its mass-normalized governing equation are

$$mL\ddot{\theta} + mg\theta = 0, \quad \ddot{\theta} + \frac{g}{L}\theta = 0.$$

Substitute $\theta = e^{st}$ and use $\ddot{\theta} = s^2 e^{st}$. Since the exponential is nonzero, its coefficient must vanish. The characteristic equation is

$$s^2 + \frac{g}{L} = 0.$$

Define the natural angular frequency $\omega_n = \sqrt{g/L}$. The roots are the purely imaginary pair $s_1 = i\omega_n$ and $s_2 = -i\omega_n$. Thus the angular solution can be written in either equivalent form

$$\theta(t) = D_1 e^{i\omega_n t} + D_2 e^{-i\omega_n t} = A_\theta \cos(\omega_n t) + B_\theta \sin(\omega_n t).$$

The frequency in hertz and oscillation period are

$$f_n = \frac{\omega_n}{2\pi}, \quad \tau = \frac{2\pi}{\omega_n}.$$

Multiplication by L gives the particle-motion solution

$$u(t) = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t} = A \cos(\omega_n t) + B \sin(\omega_n t).$$

To solve the initial value problem, evaluate both displacement and its derivative at zero. This gives $u(0) = A = u_0$ and $\dot{u}(0) = \omega_n B = v_0$, where $v_0 = \dot{u}_0$. Therefore

$$u(t) = u_0 \cos(\omega_n t) + \frac{v_0}{\omega_n} \sin(\omega_n t).$$

A displacement release sets $v_0 = 0$ and gives a cosine. A velocity release sets $u_0 = 0$ and gives a sine. Applying both initial conditions gives their sum. In exponential form the same constants are

$$C_1 = \frac{1}{2} \left(u_0 - \frac{iv_0}{\omega_n} \right), \quad C_2 = \frac{1}{2} \left(u_0 + \frac{iv_0}{\omega_n} \right).$$

Mass cancels because it multiplies both inertia and weight. A heavier ideal pendulum has the same small-angle frequency at the same length. This does not mean mass cancels from an impact calculation.

A.2 Calculate Pendulum Response to Initial Displacement

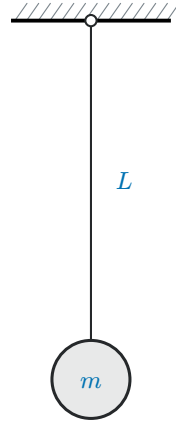


Figure 3: Pendulum of length L and mass m .

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L_{\text{nominal}} = 565 \text{ mm}$ and mass $m = 52 \text{ kg}$ as shown in Figure 3. When manufactured in series production, the pendulum lengths are not exact because of manufacturing tolerances. The lengths of the ten pendulum sets are

$$L = [561.535, 562.691, 563.845, 563.845, 565.000, \\ 565.000, 566.155, 566.155, 567.309, 568.464] \text{ mm}.$$

Do the following.

- (1) For the ten pendulum sets, calculate.
 - (a) Natural frequency ω_n in rad/s.
 - (b) Hz natural frequency f_n .
 - (c) Period of oscillation τ .
 - (d) Display in table format L, ω_n, f_n, τ .
- (2) Mean values: calculate and display the mean values of ω_n, f_n, τ .
- (3) Initial value problem (IVP). Assume initial displacement $u_0 = 2 \text{ mm}$ and zero initial velocity. Use the mean values of ω_n, f_n , and τ to complete items (a)–(c).
 - (a) Calculate and display the position of the mass $u(t_1)$ after $t_1 = 10 \text{ s}$.
 - (b) Calculate and display the duration $t_{N_{\text{osc}}}$ of N_{osc} oscillations and the corresponding position of the mass $u_{N_{\text{osc}}}$ for $N_{\text{osc}} = 10$.
 - (c) Plot the response of the system for N_{osc} oscillations and use a data tip to find $u(t_1)$ at t_1 .

Worked solution

The inputs are $g = 9.81 \text{ m/s}^2$, nominal length 565 mm, mass 52 kg, $u_0 = 2 \text{ mm}$, and $v_0 = 0$. For each actual length, calculate $\omega_{n,j} = \sqrt{g/L_j}$, then $f_{n,j} = \omega_{n,j}/(2\pi)$ and $\tau_j = 2\pi/\omega_{n,j}$.

Set	L (mm)	ω_n (rad/s)	f_n (Hz)	τ (s)
1	561.535	4.18	0.665	1.503
2	562.691	4.18	0.665	1.505
3	563.845	4.17	0.664	1.506
4	563.845	4.17	0.664	1.506
5	565.000	4.17	0.663	1.508
6	565.000	4.17	0.663	1.508
7	566.155	4.16	0.663	1.509
8	566.155	4.16	0.663	1.509
9	567.309	4.16	0.662	1.511
10	568.464	4.15	0.661	1.513

The means are arithmetic means of the individually calculated quantities. The required representative response is

$$u(t) = 0.002 \cos(\bar{\omega}_n t) \text{ m.}$$

Insert $t = 10 \text{ s}$ directly into this expression. For the assignment's duration convention, set $t_{10} = 10\bar{\tau}$ and evaluate the same response there.

Quantity	Value	Unit
$\bar{\omega}_n$	4.17	rad/s
$\bar{f}_n$	0.663	Hz
$\bar{\tau}$	1.508	s
$u(10)$	-1.352	mm
$10\bar{\tau}$	15.08	s
$u(10\bar{\tau})$	2.000	mm
$10(2\pi/\bar{\omega}_n)$	15.08	s

Because reciprocation is nonlinear, $\bar{\tau}$ is not exactly $2\pi/\bar{\omega}_n$. Consequently $10\bar{\tau}$ is not exactly ten cycles of the mean-frequency surrogate, although the difference is negligible at the requested display precision. Both durations are reported rather than silently treating the two averaging operations as identical. The surrogate is also not exactly the ensemble mean response, which would be $\frac{1}{10} \sum_j u_0 \cos(\omega_{n,j} t)$.

A2. Pendulum released from 2 mm

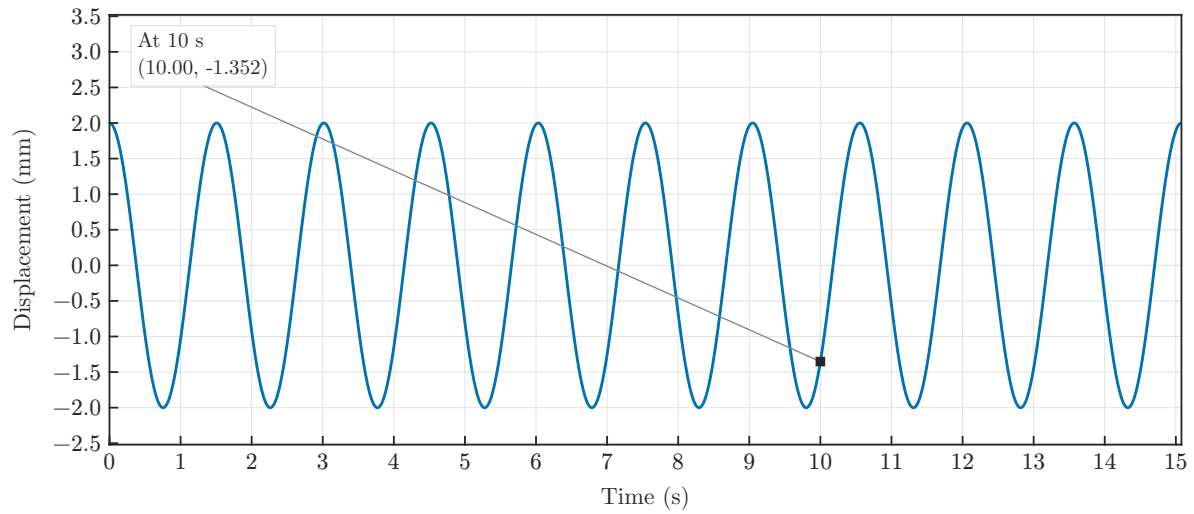


Figure 4: A.2. The square data marker evaluates the formula at exactly ten seconds.

A.3 Calculate Pendulum Response to Impact

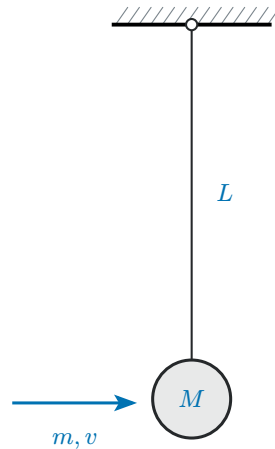


Figure 5: Impact excitation of a pendulum system.

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L = 565 \text{ mm}$ and mass $M = 52 \text{ kg}$ as shown in Figure 5. The pendulum is initially at rest hanging vertically. A small projectile of mass $m = 22 \text{ g}$ and velocity $v = 945 \text{ m/s}$ hits the mass M and remains embedded into it. This impact imparts an initial velocity to the system that starts to oscillate.

Do the following.

- (1) Display input data.
- (2) Calculate the rad/sec natural frequency ω_n and period of oscillation τ of the pendulum with projectile embedded into it.
- (3) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass.
- (4) Calculate the response $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (5) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:
 - (a) The largest response.
 - (b) The response value calculated at item (4).
- (6) Discuss your results.

Worked solution

The inputs are $L = 0.565$ m, $M = 52$ kg, projectile mass $m_p = 0.022$ kg, projectile speed $v_p = 945$ m/s, and $g = 9.81$ m/s². The target starts at rest and the projectile remains embedded.

Separate the very short collision from the subsequent vibration. During impact, conserve angular momentum about the pivot. For a tangential projectile at the bottom, the common lever arm L cancels, giving

$$m_p v_p L = (M + m_p) v_0 L, \quad v_0 = \frac{m_p v_p}{M + m_p}.$$

This is a perfectly inelastic collision. Kinetic energy is not conserved during embedding. After the collision, use the total moving mass $M + m_p$, although that mass still cancels from the ideal pendulum frequency. Since $u_0 = 0$,

$$u(t) = \frac{v_0}{\omega_n} \sin(\omega_n t), \quad u_{\max} = \frac{v_0}{\omega_n}, \quad t_{\text{first peak}} = \frac{\tau}{4}.$$

Quantity	Value	Unit
$M + m_p$	52.0	kg
v_0	0.400	m/s
ω_n	4.17	rad/s
f_n	0.663	Hz
τ	1.508	s
$u(10)$	-70.6	mm
$u_{\max}$	95.9	mm
$t_{\text{first peak}}$	0.377	s
10τ	15.08	s

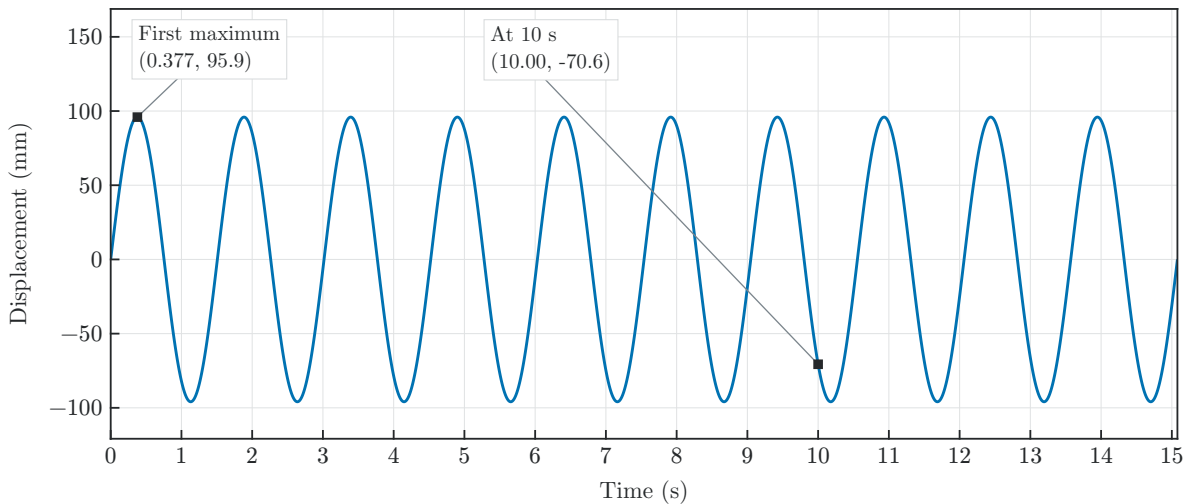
A3. Response after an embedded-projectile impact

Figure 6: A.3. Linear response following impact.

The linear amplitude corresponds to about ten degrees, so it is useful to check the small-angle approximation. After impact, conservation of mechanical energy in the exact pendulum gives

$$\frac{1}{2}v_0^2 = gL(1 - \cos \theta_{\max}).$$

Numerical integration of $\ddot{\theta} + (g/L) \sin \theta = 0$ provides an additional comparison. Both curves below use arc displacement $L\theta$, so their difference measures the linearization rather than a change of coordinate.

Quantity	Value	Unit
$\theta_{\max}$ (linear)	9.73	deg
$\theta_{\max}$ (energy)	9.74	deg
Largest arc-displacement difference over ten linear periods	10.87	mm

A3. Small-angle approximation accumulates phase error

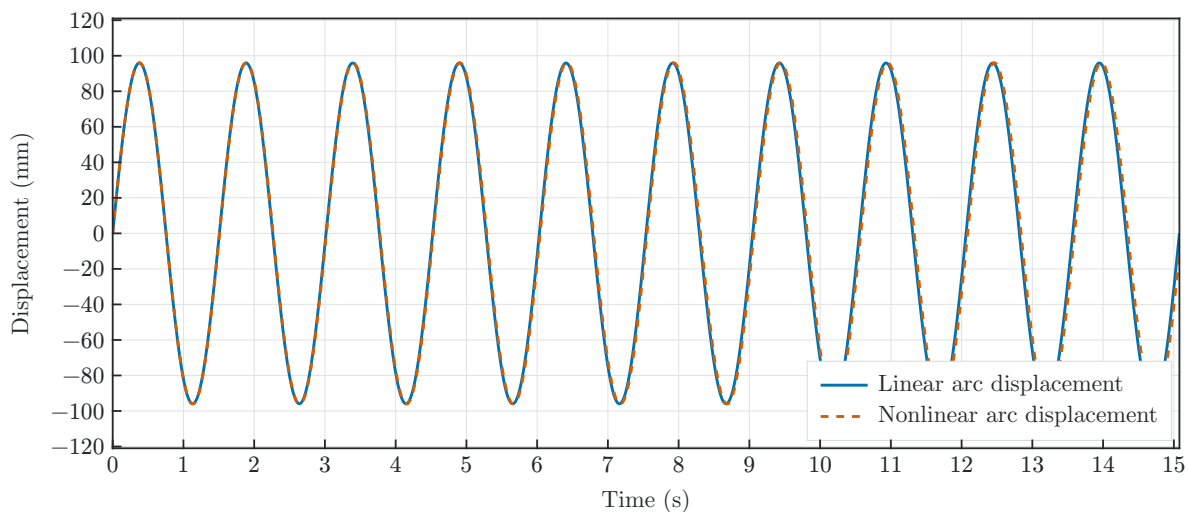


Figure 7: A.3. Small-angle and nonlinear models compared over ten linear periods.

Even a small frequency error accumulates phase error over many oscillations. The homework's linear result is the requested result; the extra comparison shows how well its assumptions work.

2 Spring-Mass-Damper Vibrating Systems

B.1 Analyze Spring-Mass Vibrating System

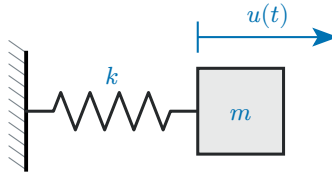


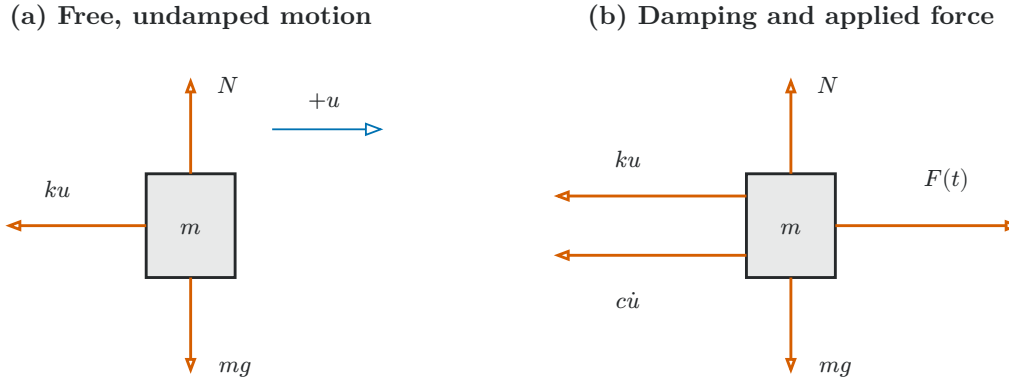
Figure 8: Spring-mass vibrating system.

Consider a spring-mass vibrating system as shown in Figure 8.

Do the following.

- (1) Draw free-body diagram (FBD).
- (2) Write NLM equation.
- (3) Deduce the equation of motion (EOM).
- (4) Deduce the governing equation. Clearly state your assumptions.
- (5) Assume $u(t) = e^{st}$ and substitute into the governing equation.
- (6) Deduce the characteristic equation.
- (7) Define the natural frequency ω_n in terms of m, k .
- (8) Solve the characteristic equation and find $s_{1,2}$ in terms of ω_n .
- (9) What type are the roots $s_{1,2}$? Choose one of the following options:
 - (a) Real.
 - (b) Imaginary.
 - (c) Complex.
- (10) Write the solution in terms of exponential functions.
- (11) Write the solution in terms of trigonometric functions.
- (12) Solve the initial value problem (IVP) and find the response $u(t)$ for:
 - (i) Initial displacement u_0 .
 - (ii) Initial velocity $\dot{u}_0$.
 - (iii) Simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$.

Worked solution



Set $c = 0$ for an undamped system; set $F = 0$ for free vibration.

For beam models, N represents the balanced vertical support reaction.

Figure 9: Free-body diagram for Problem B.1.

Assume one degree of freedom, a point mass, a massless linear spring, frictionless horizontal motion, and displacement measured from equilibrium. Weight and the vertical support reaction balance. The only horizontal force is the spring reaction $-ku$. The free-body diagram above therefore gives

$$\sum F_x = -ku = m\ddot{u}, \quad m\ddot{u} + ku = 0.$$

Divide by m and substitute $u = e^{st}$ to obtain

$$s^2 + \frac{k}{m} = 0, \quad \omega_n = \sqrt{\frac{k}{m}}, \quad s_{1,2} = \pm i\omega_n.$$

These are purely imaginary roots. The exponential and trigonometric solutions, including their initial-condition constants, are exactly those derived in A.1 with this new definition of ω_n . In particular,

$$u(t) = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t} = u_0 \cos(\omega_n t) + \frac{v_0}{\omega_n} \sin(\omega_n t).$$

The displacement-only, velocity-only, and combined cases follow by setting the unused initial condition to zero. This reuse is the main connection between the pendulum and spring problems. Both become the same normalized ordinary differential equation.

B.2 Analyze Spring-Mass-Damper Vibrating System

Consider the 1-dof damped system consisting of a spring, k , mass, m , and viscous damper, c , as presented in Figure 10. Assume the mass m is displaced from the equilibrium position by a time-dependent displacement $u(t)$.

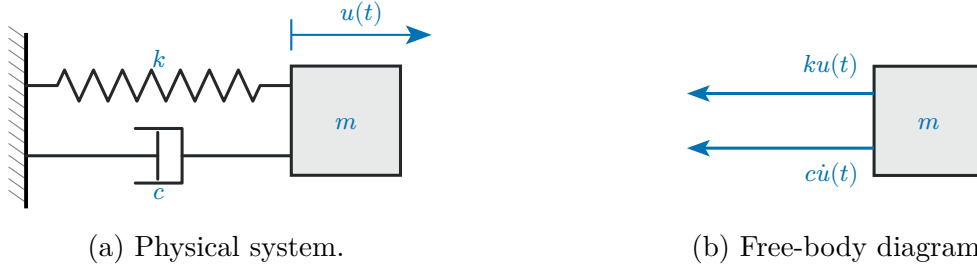


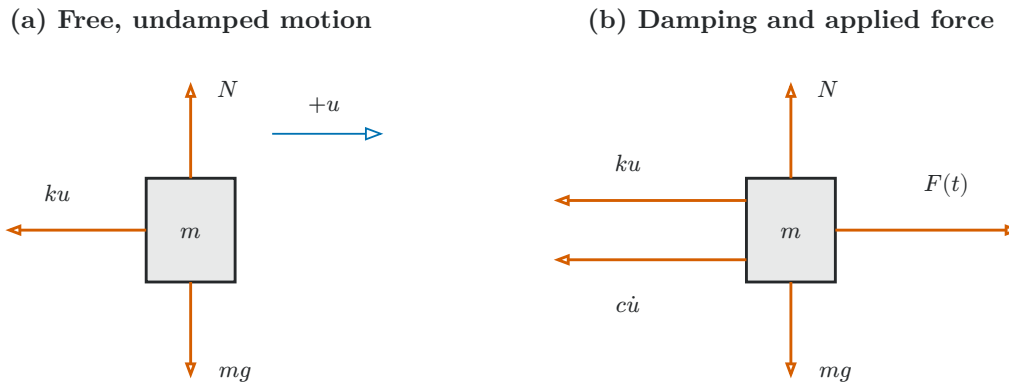
Figure 10: Mass-spring-damper vibrating system.

Do the following.

- (1) State your assumptions in performing this analysis.
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.
- (3) Write NLM equation.
- (4) Deduce the equation of motion (EOM) as an ODE in terms of m , c , k .
- (5) Assume $u(t) = e^{st}$ and substitute into EOM.
- (6) Deduce the characteristic equation.
- (7) Find the roots $s_{1,2}$ of the characteristic equation.
- (8) Define critical damping c_{cr} in terms of m and k .
- (9) Define damping ratio ζ .
- (10) Discuss the values of ζ for the three damping cases:
 - (a) Underdamped.
 - (b) Critically damped.
 - (c) Overdamped.
- (11) Express the physical damping c in terms of damping ratio ζ , mass m , and stiffness k .
- (12) Define the natural frequency ω_n in terms of m and k .
- (13) Express in terms of ω_n , ζ the $s_{1,2}$ roots found at (7).
- (14) Assume underdamped system and express the $s_{1,2}$ roots of (13) as complex numbers in terms of ω_n and ζ .
- (15) Define the damped frequency ω_d .
- (16) Substitute (15) into (14) and express the roots s_1 and s_2 in terms of ω_n , ω_d , ζ .
- (17) Assume underdamped conditions and state the inequality that ζ should satisfy.

- (18) Write the general solution of the ODE deduced at (4) in terms of constants C_1 and C_2 multiplying exponential functions of s_1 and s_2 .
- (19) Use (16) to write the solution (18) in terms of exponential functions of ω_n , ω_d , ζ .
- (20) Rewrite the solution (19) in terms of an exponential decay function multiplying the sum of two complex exponentials.
- (21) Rewrite the solution (20) in terms of an exponential decay function multiplying a trigonometric function and some constants C and ϕ to be determined from the initial conditions.
- (22) Sketch the functions discussed at (21).
- (23) Solve the initial value problem (IVP) for $\zeta < 1$ and find the response $u(t)$ for initial displacement u_0 with zero initial velocity.
- (24) Solve the initial value problem (IVP) and find the response $u(t)$ for simultaneous application of initial displacement u_0 and initial velocity $\dot{u}_0$.

Worked solution



Set $c = 0$ for an undamped system; set $F = 0$ for free vibration.

For beam models, N represents the balanced vertical support reaction.

Figure 11: Free-body diagram for Problem B.2.

Retain the B.1 assumptions and add a massless viscous damper whose force opposes relative velocity. For a fixed support, its force on the mass is $-c\dot{u}$. With no external excitation, the force balance and characteristic equation become

$$m\ddot{u} + c\dot{u} + ku = 0, \quad ms^2 + cs + k = 0.$$

The quadratic formula gives

$$s_{1,2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}.$$

Critical damping occurs when the discriminant is zero. Define

$$c_{\text{cr}} = 2\sqrt{mk} = 2m\omega_n, \quad \zeta = \frac{c}{c_{\text{cr}}}, \quad c = 2\zeta\sqrt{mk}.$$

For physical nonnegative damping, $0 < \zeta < 1$ is underdamped, $\zeta = 1$ is critically damped, and $\zeta > 1$ is overdamped. The special case $\zeta = 0$ is undamped. The normalized equation and roots are

$$\ddot{u} + 2\zeta\omega_n\dot{u} + \omega_n^2 u = 0, \quad s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}.$$

For the underdamped case define $\alpha = \zeta\omega_n$ and $\omega_d = \omega_n\sqrt{1 - \zeta^2}$. The roots are $-\alpha \pm i\omega_d$. Substitution into the exponential solution exposes its decaying envelope,

$$u(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t} = e^{-\alpha t} (C_1 e^{i\omega_d t} + C_2 e^{-i\omega_d t}) = e^{-\alpha t} [A \cos(\omega_d t) + B \sin(\omega_d t)].$$

At zero time, $u_0 = A$. Differentiating before setting time to zero gives $v_0 = -\alpha A + \omega_d B$. Hence

$$A = u_0, \quad B = \frac{v_0 + \alpha u_0}{\omega_d},$$

and the complete initial-value solution is

$$u(t) = e^{-\alpha t} \left[u_0 \cos(\omega_d t) + \frac{v_0 + \alpha u_0}{\omega_d} \sin(\omega_d t) \right].$$

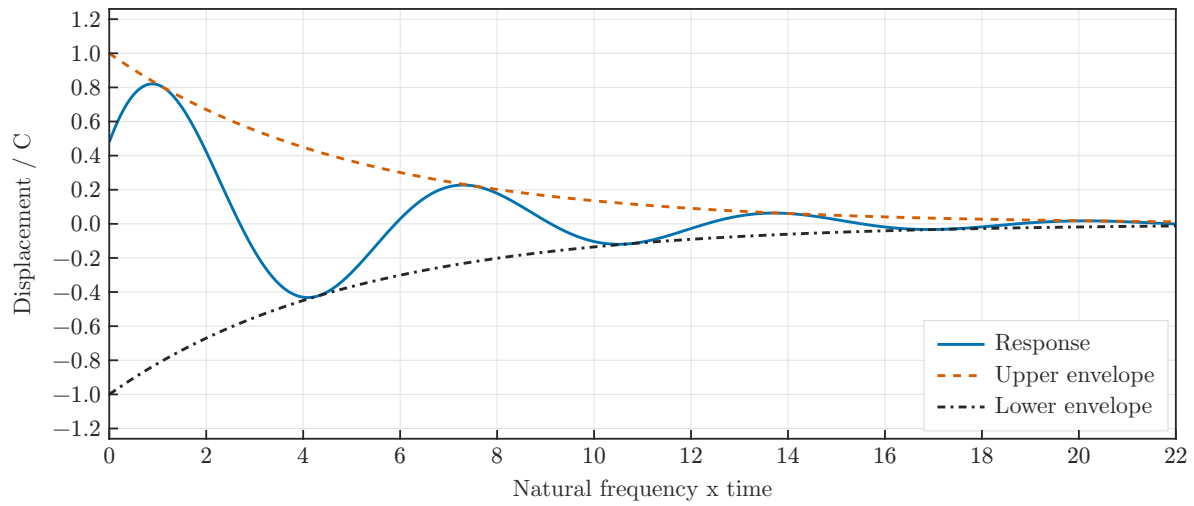
For a displacement release, set $v_0 = 0$. The sine term remains necessary to make the initial velocity zero. Simply multiplying $u_0 \cos(\omega_d t)$ by an exponential gives the wrong initial derivative.

To convert forms, use

$$C_1 = \frac{A - iB}{2}, \quad C_2 = \frac{A + iB}{2}, \quad C = \sqrt{A^2 + B^2}, \quad \phi = \text{atan2}(A, B).$$

Then $u(t) = C e^{-\alpha t} \sin(\omega_d t + \phi)$. Sketch a sinusoid bounded by $\pm C e^{-\alpha t}$. Its successive peaks shrink while the damped period $\tau_d = 2\pi/\omega_d$ remains constant. The response plot in B.3 illustrates this behavior. The envelope intercept C need not equal an attained displacement peak.

The following sketch separates oscillation from its exponential envelope. It illustrates the general form using $\omega_n = 1$, $\zeta = 0.20$, and phase $\phi = 0.5$ radians; these are illustrative parameters, not additional assigned data.

B2. Underdamped response and exponential envelope**Figure 12:** *B.2. Oscillation inside the exponential decay envelope.*

B.3 Calculate Spring-Mass-Damper Response to Initial Displacement

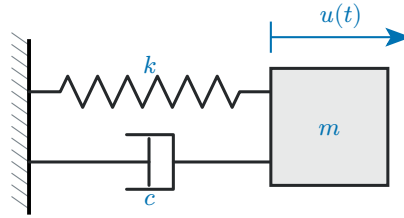


Figure 13: *Spring-mass-damper vibrating system.*

Consider a spring-mass vibrating system with $m = 12 \text{ kg}$, $k = 655 \text{ N/m}$, $c = 19 \text{ N/(m/s)}$ as shown in Figure 13. The vibrating system is given an initial displacement $u_0 = 3.5 \text{ mm}$ with zero initial velocity.

Do the following.

- (1) Display input data.
- (2) Calculate the natural frequency in rad/sec ω_n and in Hz f_n .
- (3) Calculate the critical damping c_{cr} .
- (4) Calculate the damping ratio ζ .
- (5) Calculate the damped frequency in rad/sec ω_d and in Hz f_d .
- (6) Calculate the damped period of oscillation τ_d .
- (7) Calculate the characteristic-equation roots s_1 , s_2 and the constants C_1 , C_2 .
- (8) Calculate the constants A , B and then the constants C , ϕ .
- (9) Calculate the duration t_1 of one oscillation and the corresponding position $u_1 = u(t_1)$.
- (10) Calculate the value $u_{0.02}$ representing 2% of the initial displacement.
- (11) Plot the response of the system for $N_{osc} = 10$ oscillations and use datatips on the plot to find the following:
 - (a) The largest response.
 - (b) The response value calculated at item (9).
 - (c) The time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (10).
- (12) Discuss your results.

Worked solution

The inputs are $m = 12$ kg, $k = 655$ N/m, $c = 19$ N s/m, $u_0 = 3.5$ mm, and $v_0 = 0$. Calculate ω_n , c_{cr} , ζ , and ω_d in that order. Insert them into the B.2 solution. The table includes both exponential and trigonometric constants.

Quantity	Value	Unit
k	655	N/m
ω_n	7.39	rad/s
f_n	1.176	Hz
c_{cr}	177.3	N s/m
ζ	0.1072	1
c	19.00	N s/m
ω_d	7.35	rad/s
f_d	1.169	Hz
τ_d	0.855	s
$\text{Re}(s_{1,2})$	-0.792	1/s
$ \text{Im}(s_{1,2}) $	7.35	1/s
A	3.50	mm
B	0.377	mm
C	3.52	mm
ϕ (sine convention)	1.463	rad
$\text{Re}(C_1) = \text{Re}(C_2)$	1.750	mm
$\text{Im}(C_1) = -\text{Im}(C_2)$	-0.1886	mm
$u(\tau_d)$	1.778	mm
$0.02u_0$	0.0700	mm
t_s (last crossing)	4.79	s
t_{envelope}	4.95	s

For one complete damped cycle, $\cos(\omega_d \tau_d) = 1$ and $\sin(\omega_d \tau_d) = 0$, so

$$u(\tau_d) = u_0 e^{-\alpha \tau_d}.$$

For this displacement-release problem, the largest absolute displacement is the initial value. The subsequent stationary points occur at $t_n = n\pi/\omega_d$ with decreasing magnitudes $|u_0|e^{-\alpha t_n}$.

The 2% settling time means the earliest time after which every future displacement satisfies $|u(t)| \leq 0.02|u_0|$. It is not the first time the oscillation enters that band, and it is not a zero crossing. Locate the last extremum above the band, then solve the falling side of that lobe for the exact final crossing. Because all later extrema are smaller, this proves the response stays inside the band.

An easier sufficient bound comes from the exponential envelope. For this zero-velocity release,

$$t_{\text{envelope}} = \frac{1}{\alpha} \ln \left(\frac{1}{0.02\sqrt{1-\zeta^2}} \right).$$

The familiar approximation $4/(\zeta\omega_n)$ is not an exact last-crossing calculation. Both the exact result and the envelope bound appear in the table.

B3. Displacement release

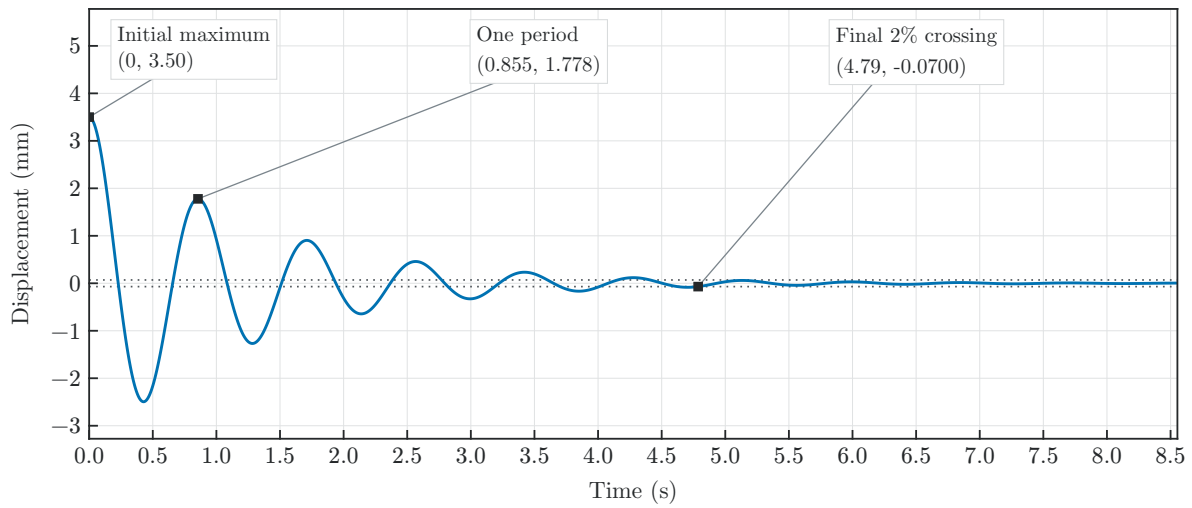


Figure 14: B.3. Exact final crossing and the 2% band. The detail plot resolves the final crossing of the settling band.

B3. Final crossing of the 2% band

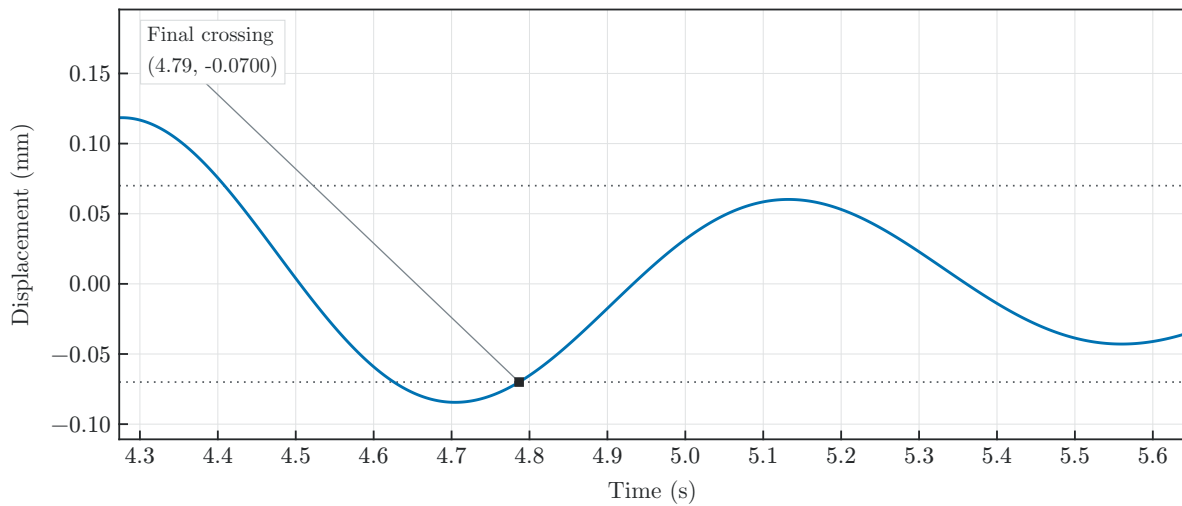


Figure 15: B.3. Detail of the final 2% crossing.

B.4 Calculate Truck Suspension Response

Consider a pickup truck wheel suspension consisting of a leaf spring and an oil damper. The truck and its suspension can be modeled, in first approximation, as a k - m - c spring-mass-damper system as shown in Figure 16.

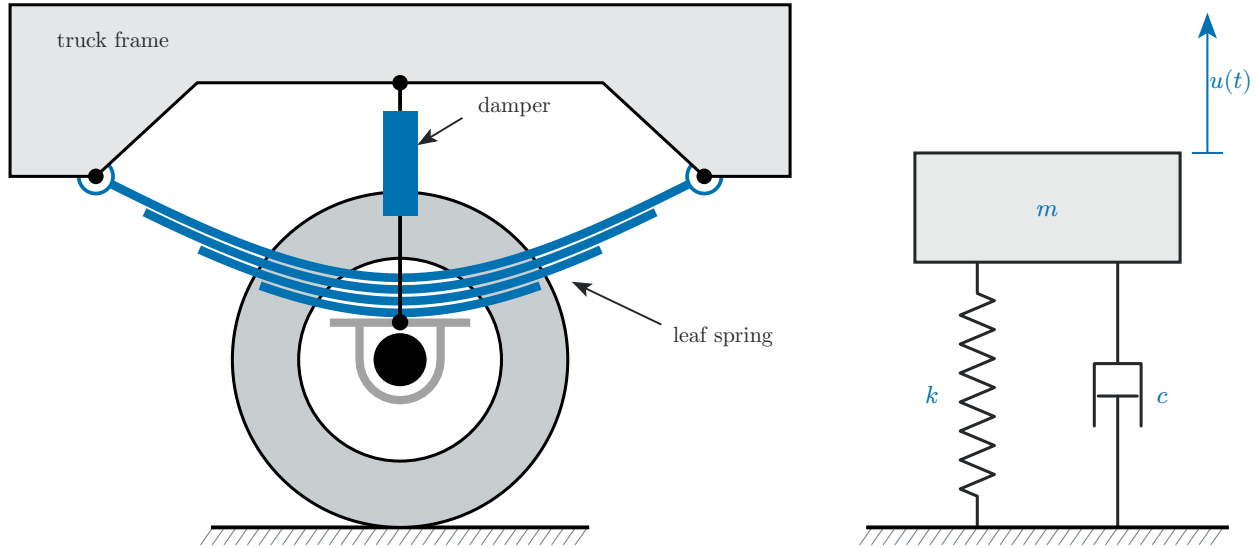


Figure 16: Truck suspension consisting of a leaf spring and an oil damper. The wheel contact is treated as the road datum in the one-degree-of-freedom idealization.

The leaf spring can be approximated by a uniform beam of length $L = 830 \text{ mm}$ and cross-section dimensions $h = 3 \text{ mm}$ and $b = 65 \text{ mm}$, made of steel with $E = 200 \text{ GPa}$. The truck mass apportioned to the wheel can be estimated as $m = 457 \text{ kg}$. Assume initial displacement $u_0 = 125 \text{ mm}$ and zero initial velocity $\dot{u}_0 = 0$.

Do the following.

- (1) Display input data.
- (2) Calculate the second moment of area I of the beam cross section.
- (3) Calculate the flexural stiffness EI of the beam.
- (4) Calculate the spring constant k of the system.
- (5) Calculate the natural frequency in rad/sec ω_n and in Hz f_n .
- (6) Calculate the critical damping c_{cr} .
- (7) Assume a 25% damping ratio, i.e., $\zeta = 0.25$, and calculate the physical damping c . Print ζ and c .
- (8) Calculate the damped frequency in rad/sec ω_d and in Hz f_d .
- (9) Calculate the damped period of oscillation τ_d .
- (10) Calculate the characteristic-equation roots s_1, s_2 and the constants C_1, C_2 .
- (11) Calculate the constants A, B and then the constants C, ϕ .

- (12) Calculate the duration t_1 of one oscillation and the corresponding position $u_1 = u(t_1)$.
- (13) Calculate the value $u_{0.02}$ representing 2% of the initial displacement.
- (14) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and use datatips on the plot to find the following:
- (a) The largest response.
 - (b) The response value calculated at item (12).
 - (c) The time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (13).
- (15) Discuss your results.
- (16) Graduate work. Assume the damper is worn out such that ζ is reduced to $\zeta = 10\%$.
- (a) Repeat the calculation of items (6) through (12).
 - (b) Compare and discuss the two results.

Worked solution

The inputs are $L = 0.830$ m, $b = 0.065$ m, $h = 0.003$ m, $E = 200$ GPa, $m = 457$ kg, $u_0 = 0.125$ m, and $v_0 = 0$. Use $\zeta = 0.25$ for the initial case and $\zeta = 0.10$ for the worn damper.

Treat the leaf spring as the equivalent simply supported beam with a center load and full support span L . This is the boundary-condition assumption consistent with the spring-eye supports and the note formula on pages 249 and 251. Its center deflection under load P is $PL^3/(48EI)$, so

$$I = \frac{bh^3}{12}, \quad EI = EI, \quad k = \frac{P}{\delta} = \frac{48EI}{L^3}.$$

Do not use the cantilever factor of three here. The stiffness depends on the support conditions as well as the material and section dimensions.

Quantity	Value	Unit
I	146.2	mm ⁴
EI	29.3	N m ²
k	2.46e+03	N/m
mg/k using $g = 9.81$	1.826	m

Now use the same B.2 formulas as before. For the original damper the numerical results are

Quantity	Value	Unit
k	2.46e+03	N/m
ω_n	2.32	rad/s
f_n	0.369	Hz
c_{cr}	2.12e+03	N s/m
ζ	0.250	1
c	530	N s/m
ω_d	2.24	rad/s
f_d	0.357	Hz
τ_d	2.80	s
$\text{Re}(s_{1,2})$	-0.579	1/s
$ \text{Im}(s_{1,2}) $	2.24	1/s
A	125.0	mm
B	32.3	mm
C	129.1	mm
ϕ (sine convention)	1.318	rad
$\text{Re}(C_1) = \text{Re}(C_2)$	62.5	mm
$\text{Im}(C_1) = -\text{Im}(C_2)$	-16.14	mm
$u(\tau_d)$	24.7	mm
$0.02u_0$	2.50	mm
t_s (last crossing)	6.09	s
t_{envelope}	6.81	s

B4 healthy. Displacement release

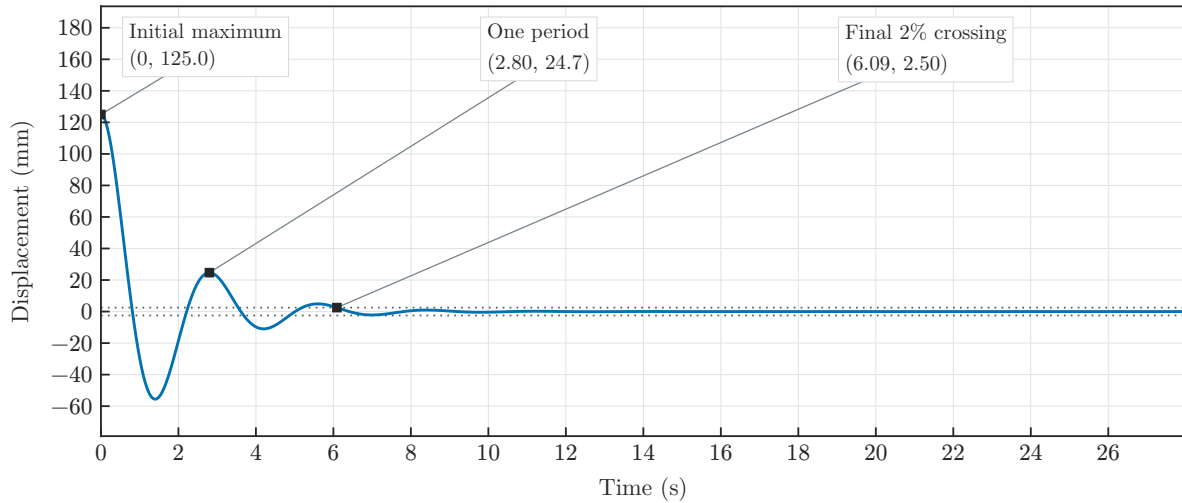


Figure 17: *B.4. Original damper, with the required response markers.*

For the worn damper the spring and mass are unchanged. Therefore ω_n and c_{cr} are unchanged, while c , ω_d , the roots, the constants, and the decay change.

Quantity	Value	Unit
k	2.46e+03	N/m
ω_n	2.32	rad/s
f_n	0.369	Hz
c_{cr}	2.12e+03	N s/m
ζ	0.1000	1
c	212	N s/m
ω_d	2.31	rad/s
f_d	0.367	Hz
τ_d	2.72	s
$\text{Re}(s_{1,2})$	-0.232	1/s
$ \text{Im}(s_{1,2}) $	2.31	1/s
A	125.0	mm
B	12.56	mm
C	125.6	mm
ϕ (sine convention)	1.471	rad
$\text{Re}(C_1) = \text{Re}(C_2)$	62.5	mm
$\text{Im}(C_1) = -\text{Im}(C_2)$	-6.28	mm
$u(\tau_d)$	66.5	mm
$0.02u_0$	2.50	mm
t_s (last crossing)	16.56	s
t_{envelope}	16.90	s

B4 worn. Displacement release

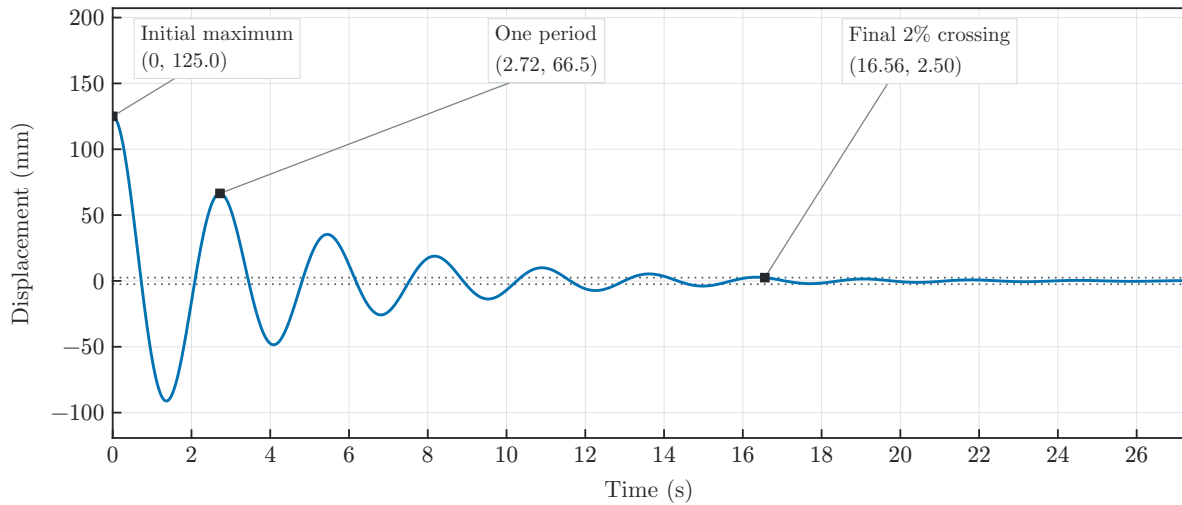


Figure 18: *B.4. Worn damper, with the required response markers.*

B4. Worn damping increases the duration of motion

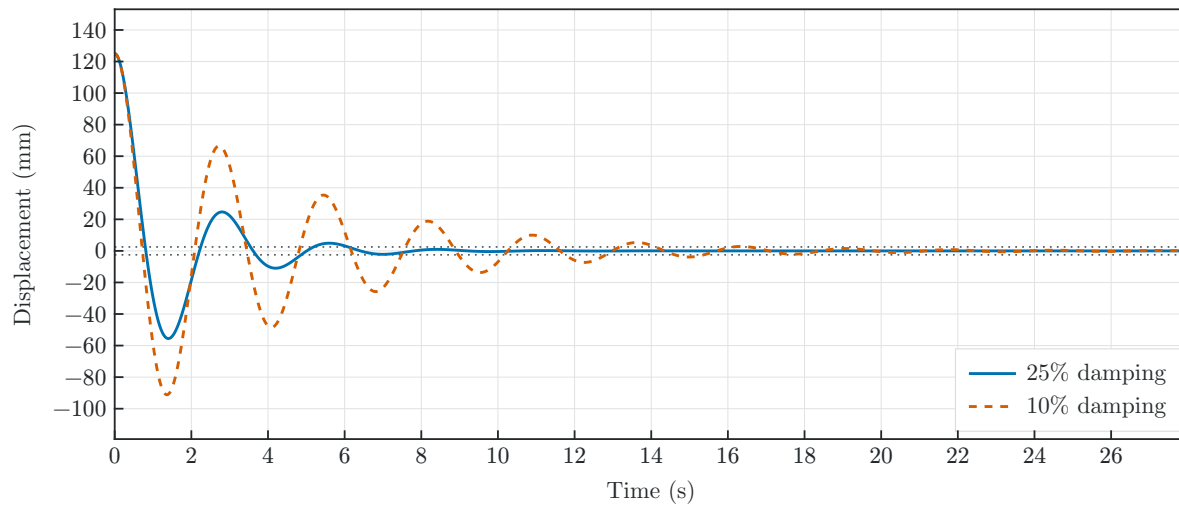


Figure 19: B.4. Displacement responses using identical initial conditions.

The worn damper permits more oscillation before settling. Its damped period is slightly shorter because damping lowers oscillation frequency. Both cases begin at the same largest displacement, so the relevant comparison is persistence and subsequent extrema rather than the initial peak.

The given 3 mm uniform beam predicts a very large static deflection mg/k and a large displacement relative to its span. These are formal linear-model results. A physical suspension calculation would require the actual leaf stack, geometry, stiffness, and available travel.

B.5 Overdamped Spring-Mass-Damper Systems

Consider a spring-mass-damper system with natural frequency $f_n = 6.25$ Hz and various values of the damping ratio ζ . The system is given an initial displacement $u_0 = 13$ mm with zero initial velocity.

Do the following.

- (1) Display input data.
- (2) Calculate the rad/s frequency ω_n .
- (3) Assume overdamped conditions $\zeta > 1$ and develop the solution as follows:
 - (a) Write the expressions of the roots s_1, s_2 in terms of ω_n and ζ .
 - (b) Write the solution $u(t)$ in terms of s_1, s_2 and constants C_1, C_2 .
 - (c) Write the expression of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$.
 - (d) Take $\zeta = 2.3$ and calculate the numerical values of s_1, s_2, C_1 , and C_2 .
 - (e) Plot the response $u(t)$ for $t = 0, \dots, 3$ s.
- (4) Assume critically damped conditions $\zeta = 1$ and complete items (a)–(e).
 - (a) Write the expressions of the two partial solutions $u_1(t), u_2(t)$ in terms of ω_n .
 - (b) Write the expression of the complete solution in terms of ω_n and C_1, C_2 .
 - (c) Write the expressions of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$.
 - (d) Calculate the numerical values of C_1, C_2 .
 - (e) Plot the response $u(t)$ for $t = 0, \dots, 3$ s.
- (5) Graduate work. Derive the ODE for $\zeta = 1$ and verify that the two solutions $u_1(t)$ and $u_2(t)$ from (4)(a) satisfy the ODE.
- (6) Graduate work. Some race-car manufacturers design for critical damping, whereas others use an overdamped design. Discuss the advantages and disadvantages of the two options.

Worked solution

The inputs are $f_n = 6.25$ Hz, $u_0 = 13$ mm, $v_0 = 0$, and a time interval from zero to three seconds. First calculate $\omega_n = 2\pi f_n$.

For $\zeta > 1$, the roots are distinct negative real numbers,

$$s_{1,2} = -\omega_n \left(\zeta \mp \sqrt{\zeta^2 - 1} \right).$$

The response is $u = C_1 e^{s_1 t} + C_2 e^{s_2 t}$. Apply $C_1 + C_2 = u_0$ and $s_1 C_1 + s_2 C_2 = v_0$ to obtain

$$C_1 = \frac{v_0 - s_2 u_0}{s_1 - s_2}, \quad C_2 = \frac{s_1 u_0 - v_0}{s_1 - s_2}.$$

For $\zeta = 1$, the roots coincide at $-\omega_n$. Two independent solutions are $u_1 = e^{-\omega_n t}$ and $u_2 = t e^{-\omega_n t}$. Their linear combination and initial-condition constants are

$$u(t) = (C_1 + C_2 t) e^{-\omega_n t}, \quad C_1 = u_0, \quad C_2 = v_0 + \omega_n u_0.$$

The numerical constants for $\zeta = 2.3$ and $\zeta = 1$ are

Quantity	Value	Unit
ω_n	39.3	rad/s
s_1 (overdamped)	-8.98	1/s
s_2 (overdamped)	-171.7	1/s
C_1 (overdamped)	13.72	mm
C_2 (overdamped)	-0.718	mm
C_1 (critical)	13.00	mm
C_2 (critical)	511	mm/s

For the graduate verification, the critical equation is $(D + \omega_n)^2 u = 0$, where $D = d/dt$. The first solution satisfies $(D + \omega_n) e^{-\omega_n t} = 0$. The second satisfies $(D + \omega_n)(t e^{-\omega_n t}) = e^{-\omega_n t}$, and a second application gives zero. Explicitly, the derivatives of the second solution are

$$\dot{u}_2 = (1 - \omega_n t) e^{-\omega_n t}, \quad \ddot{u}_2 = (-2\omega_n + \omega_n^2 t) e^{-\omega_n t},$$

which cancel on substitution into $\ddot{u} + 2\omega_n \dot{u} + \omega_n^2 u = 0$.

B5. Critical and overdamped displacement release

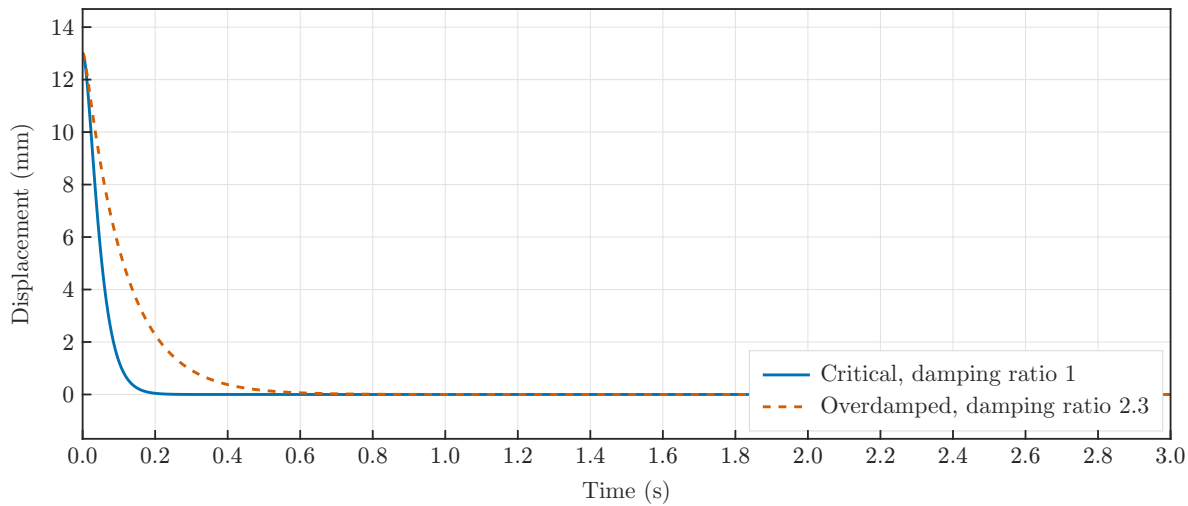


Figure 20: B.5. Required zero-to-three-second interval.

B5. Critical and overdamped displacement release

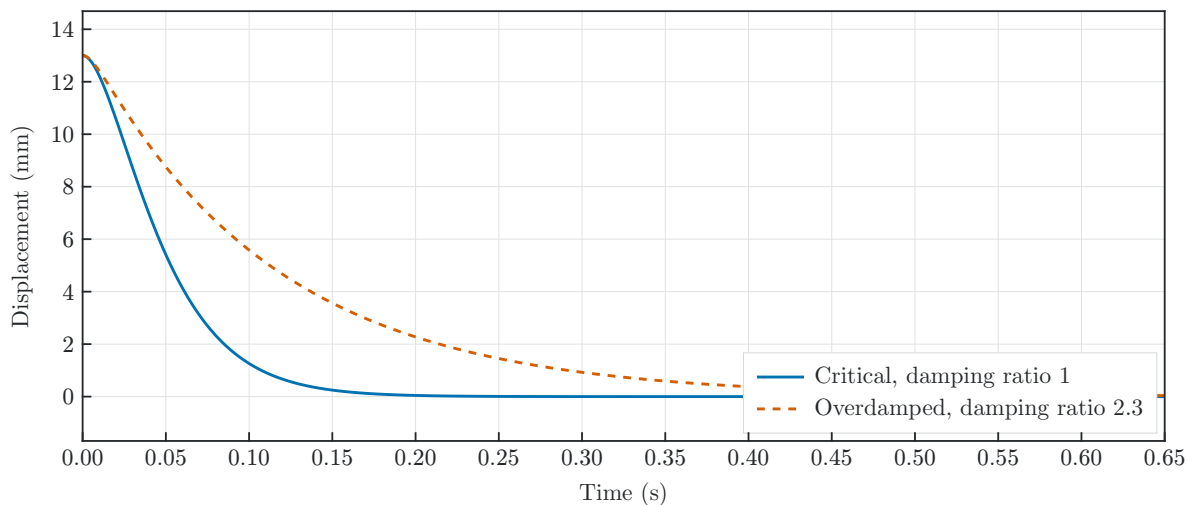


Figure 21: B.5. A shorter view resolves the initial behavior.

For this fixed mass and stiffness with an initial displacement and zero velocity, critical damping returns to equilibrium most rapidly among the nonoscillatory cases. Overdamping reduces the early displacement more slowly and introduces a long decay associated with the less-negative root. Neither response oscillates for these initial conditions.

For the race-car discussion, critical damping favors quick recovery without oscillatory overshoot. Overdamping supplies greater dissipation but recovers more slowly and transmits larger damper forces for a given relative velocity. Choosing between them for a vehicle also requires road, tire, wheel, and suspension-travel models.

B.6 Analyze Spring-Mass-Damper Forced Vibration

Consider a mass-spring-damper vibrating system excited by a time-dependent force $F(t)$ as shown in Figure 22.

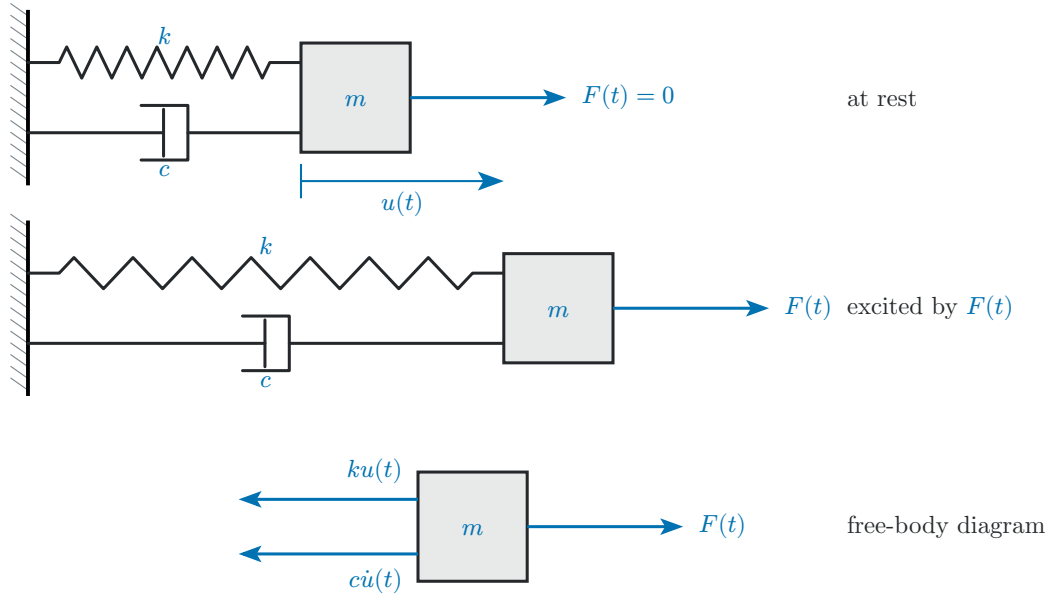


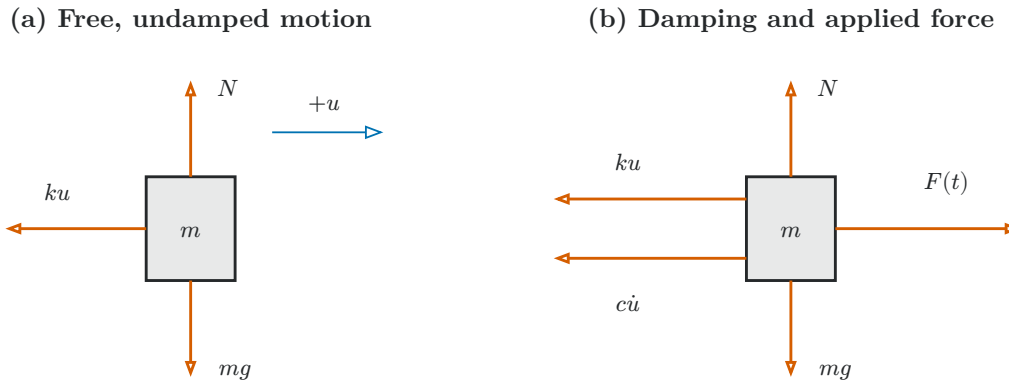
Figure 22: Spring-mass-damper system excited by force $F(t)$.

Do the following.

- (1) State your assumptions in performing this analysis.
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.
- (3) Write NLM equation.
- (4) Deduce the equation of motion (EOM) as an ODE in terms of m , c , k , $F(t)$.
- (5) Assume harmonic excitation $F(t) = \hat{F} e^{i\omega t}$, substitute in (4), and divide by m .
- (6) Define the normalized excitation amplitude $\hat{f} = \hat{F}/m$ and recall the definitions of ω_n and ζ to obtain a nonhomogeneous linear ODE in ω_n , ζ , $\hat{f}$.
- (7) Write the general solution $u(t)$ as a sum of a complementary solution $u_c(t)$ and a particular solution $u_p(t)$.
- (8) Write the general form of the complementary solution $u_c(t)$.
- (9) Find the particular solution $u_p(t)$.
- (10) Use (8), (9) to write the complete solution $u(t)$; discuss the meaning of transient response and steady-state response.
- (11) Find an expression for the steady-state response amplitude $\hat{u}_{ss}$ in terms of ω_n , ζ , $\hat{f}$, and ω .
- (12) Develop the expression for the frequency response function $\text{FRF}(\omega)$ in terms of ω_n , ζ , ω .
- (13) Discuss the variation with frequency of FRF magnitude and phase.

- (14) Give the definition of the magnification factor $M(\omega)$.
- (15) Give the expression for the magnification factor $M(\omega)$ in terms of ω_n and ζ .
- (16) Give the definition of the resonance frequency ω_r .
- (17) Give the expression for the resonance frequency ω_r in terms of ω_n and ζ .
- (18) Give the expression of the peak magnification M_r .
- (19) Give the expression for the FRF phase $\phi(p)$, where $p = \omega/\omega_n$.
- (20) Give the definition of phase resonance and deduce the FRF expression at phase resonance.
- (21) Calculate the magnification factor at phase resonance.
- (22) Calculate the magnitude of the physical response at phase resonance.
- (23) Discuss the difference between true resonance and phase resonance.
- (24) Write and compare the expressions for the resonance frequency ω_r and damped frequency ω_d .
- (25) Establish a formula to calculate the RPM corresponding to a rad/sec frequency ω .

Worked solution



Set $c = 0$ for an undamped system; set $F = 0$ for free vibration.

For beam models, N represents the balanced vertical support reaction.

Figure 23: Free-body diagram for Problem B.6.

Assume a fixed support, linear spring, viscous damping, a point mass, small motion, and prescribed force $F(t)$. The horizontal free-body diagram gives $F(t) - c\dot{u} - ku = m\ddot{u}$. For harmonic forcing represented by a complex exponential,

$$m\ddot{u} + c\dot{u} + ku = \hat{F}e^{i\omega t}.$$

Physical force and displacement are the real parts. Define $\hat{f} = \hat{F}/m$ and divide by m to obtain

$$\ddot{u} + 2\zeta\omega_n\dot{u} + \omega_n^2u = \hat{f}e^{i\omega t}.$$

The complete response is $u = u_c + u_p$. For an underdamped system, $u_c = e^{-\alpha t}[A \cos(\omega_d t) + B \sin(\omega_d t)]$. Seek a particular solution $u_p = \hat{u}e^{i\omega t}$. Substituting its derivatives gives

$$(-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2)\hat{u} = \hat{f},$$

so

$$\hat{u} = \frac{\hat{f}}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega} = \frac{\hat{F}}{k - m\omega^2 + ic\omega}.$$

For positive damping, the complementary response decays and is called the transient. The particular response persists at the excitation frequency and is called the steady-state response. To satisfy initial conditions, apply them to the complete solution, not to the transient alone.

Let $u_{\text{qst}} = \hat{F}/k$. The course defines the dimensionless frequency response function (FRF) normalized by this quasi-static displacement,

$$\text{FRF}(\omega) = \frac{\hat{u}}{u_{\text{qst}}} = \frac{\omega_n^2}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega} = \frac{1}{1 - p^2 + 2i\zeta p}, \quad p = \frac{\omega}{\omega_n}.$$

The physical compliance $H = \hat{u}/\hat{F} = \text{FRF}/k$ is a different quantity with units of metres per newton. The distinction prevents unit errors in B.7.

The magnification factor and phase are

$$M(p) = |\text{FRF}| = \frac{1}{\sqrt{(1 - p^2)^2 + (2\zeta p)^2}}, \quad \phi(p) = -\text{atan2}(2\zeta p, 1 - p^2).$$

Thus $|\hat{u}| = (\hat{F}/k)M$ and the real steady-state response to $\hat{F} \cos \omega t$ is $|\hat{u}| \cos(\omega t + \phi)$. At very low frequency, $M \rightarrow 1$ and phase approaches zero. At large frequency, $M \sim 1/p^2$ and phase approaches -180° .

To find an interior amplitude peak, minimize the denominator squared,

$$Q(p) = (1 - p^2)^2 + 4\zeta^2 p^2, \quad \frac{dQ}{dp} = 4p(p^2 - 1 + 2\zeta^2).$$

For $0 < \zeta < 1/\sqrt{2}$, the nonzero stationary point gives

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2}, \quad M_r = \frac{1}{2\zeta \sqrt{1 - \zeta^2}}.$$

For $\zeta \geq 1/\sqrt{2}$, the maximum occurs at zero frequency and equals one. For zero damping, the response is unbounded at exact resonance within this ideal model.

Phase resonance means that displacement lags force by 90° . It occurs at $p = 1$ for positive damping. At this frequency,

$$\text{FRF}(\omega_n) = \frac{1}{2i\zeta} = -\frac{i}{2\zeta}, \quad M_{90} = \frac{1}{2\zeta}, \quad \hat{u}_{90} = -i\frac{\hat{F}}{2\zeta k}.$$

True amplitude resonance is a peak of the magnitude under constant force amplitude, whereas phase resonance is the quadrature condition. In the range with a finite nonzero peak,

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2} < \omega_d = \omega_n \sqrt{1 - \zeta^2} < \omega_n.$$

Finally, with one excitation cycle per revolution, the rotational speed is $N = 60\omega/(2\pi)$ revolutions per minute, abbreviated rpm.

B.7 Calculate Flex-Beam Response to Slightly Unbalanced Electric Motor Excitation

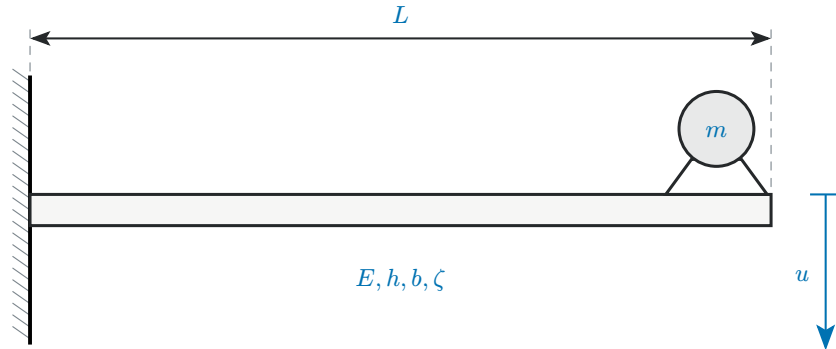


Figure 24: Damped cantilever beam excited by force $F(t)$ from slightly unbalanced electric motor.

Consider a cantilever beam of length $L = 568 \text{ mm}$, elastic modulus $E = 200 \text{ GPa}$, cross-section dimensions $h = 3 \text{ mm}$ and $b = 37 \text{ mm}$, and damping ratio $\zeta = 14\%$. (The damping is achieved through a proprietary damping treatment applied on the beam surface.) An electric motor of mass $m = 0.52 \text{ kg}$ operating at $1,500 \text{ RPM}$ is attached at the beam tip as shown in Figure 24. The electric motor is slightly unbalanced, producing a slight vibratory excitation $F(t) = \hat{F}e^{i\omega t}$ with $\hat{F} = 12 \text{ mN}$. The excitation frequency ω depends on the electric motor speed.

Do the following.

- (1) Display input data.
- (2) Calculate:
 - (a) Cross-section second moment of area I , flexural stiffness EI , and spring constant k .
 - (b) Natural frequency ω_n rad/s, critical damping c_{cr} , physical damping c .
 - (c) Quasi-static deflection u_{qst} corresponding to $\omega_n \rightarrow 0$.
- (3) Calculate the values of:
 - (a) Resonant frequency ω_r and damped frequency ω_d .
 - (b) Peak magnification M_r .
 - (c) Magnification at phase resonance M_{90} .
 - (d) Physical response $\hat{u}_{90}$ at phase resonance and its magnitude $|\hat{u}_{90}|$.
 - (e) Calculate the Hz frequencies f_r , f_n and the corresponding RPM values.
- (4) Assume excitation at phase resonance:
 - (a) Calculate the excitation frequency ω in rad/s and f in Hz.
 - (b) Calculate period of oscillation τ .
 - (c) Plot the time response for $N_{osc} = 10$ oscillations. Hint: plot the real part of the complex displacement $u(t)$.
- (5) Graduate work. Complete items (a)–(d).

- (a) Calculate and display the operational frequency f_0 in Hz and the corresponding ω_0 in rad/sec. Plot the magnitude and phase of the frequency response function FRF vs. excitation f in Hz from quasi-static to f_0 with $N_f = 5000$ points.
- (b) Calculate and display the FRF magnitude at phase-resonance RPM, at double the phase resonance RPM, and at operational RPM. Display the corresponding Hz frequencies. Put the corresponding data tips on the magnitude plot.
- (c) Calculate the vibration magnitude $|u_0|$ when the electric motor runs at its operational RPM.
- (d) Comment on what vibration would be observed as the motor is switched on and accelerates to reach its operational RPM.

Worked solution

The inputs are $L = 0.568$ m, $h = 0.003$ m, $b = 0.037$ m, $E = 200$ GPa, motor mass $m = 0.52$ kg, $\zeta = 0.14$, $\hat{F} = 0.012$ N, and operating speed 1500 rpm. Neglect beam mass and use the weak-axis vertical bending shown in the original figure. This lumped approximation is prescribed here; a distributed beam model would need additional mass information.

The quasi-static limit in item (2c) is $\omega \rightarrow 0$ with ω_n fixed. The printed $\omega_n \rightarrow 0$ is a typographical error.

Compute $I = bh^3/12$, $k = 3EI/L^3$, $\omega_n = \sqrt{k/m}$, and $c = 2\zeta m\omega_n$. The force-normalized limit in B.6 then gives $u_{\text{qst}} = \hat{F}/k$.

Quantity	Value	Unit
I	83.2	mm ⁴
EI	16.65	N m ²
k	273	N/m
ω_n	22.9	rad/s
f_n	3.64	Hz
c_{cr}	23.8	N s/m
c	3.33	N s/m
u_{qst}	44.0	μm
ω_r	22.4	rad/s
ω_d	22.7	rad/s
f_r	3.57	Hz
N_r	214	rpm
N_{90}	219	rpm
M_r	3.61	1
M_{90}	3.57	1
$ \hat{u}_{90} $	157.2	μm
τ_{90}	0.274	s
f_0	25.0	Hz
ω_0	157.1	rad/s
$ \hat{u}(f_0) $	0.955	μm

At phase resonance use $\omega = \omega_n$, $f = f_n$, and $\tau = 1/f_n$. For real force $\hat{F} \cos(\omega_n t)$, the steady-state physical displacement is

$$u_{ss}(t) = \frac{u_{qst}}{2\zeta} \sin(\omega_n t).$$

Equivalently, the complex response amplitude is $\hat{u}_{90} = -i|\hat{u}_{90}|$. The assignment does not prescribe starting conditions for the phase-resonant plot, so the steady-state curve is the direct interpretation. The second curve shows a force switched on at time zero from rest. In that case the transient must cancel the initial steady-state velocity, giving

$$u_{rest}(t) = U \sin(\omega_n t) - \frac{U\omega_n}{\omega_d} e^{-\zeta\omega_n t} \sin(\omega_d t), \quad U = \frac{u_{qst}}{2\zeta}.$$

B7. Phase-resonant response

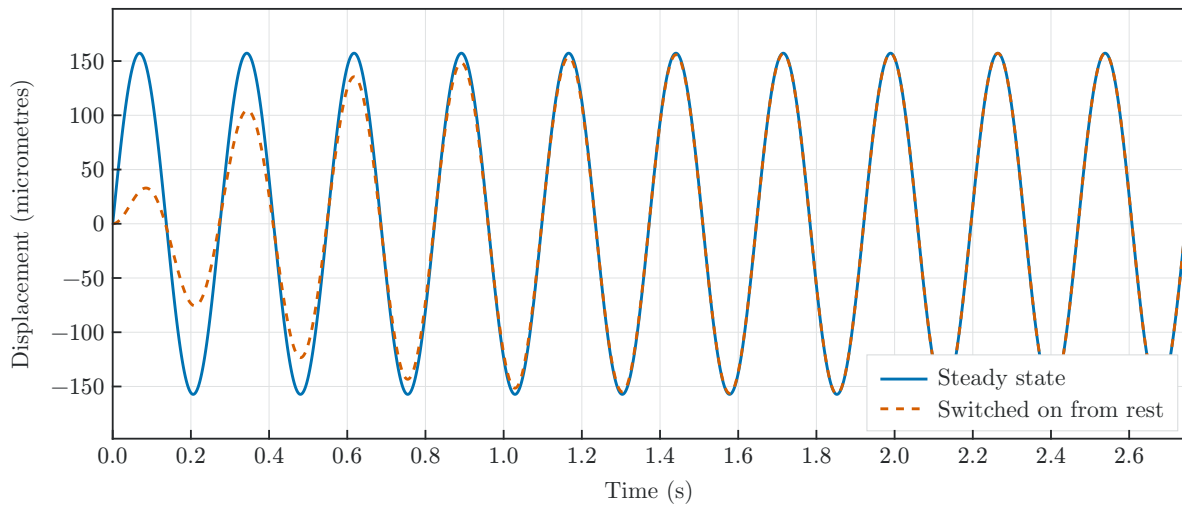


Figure 25: B.7. Steady-state response and a separately identified rest-start interpretation.

The requested frequency sweep uses exactly 5000 uniformly spaced points from zero to the operational frequency $f_0 = 25$ Hz. Data markers at the requested frequencies are calculated analytically rather than substituted by the nearest sweep sample.

Quantity	Value	Unit
Phase resonance f	3.64	Hz
Double resonance RPM f	7.29	Hz
Operating RPM f	25.0	Hz
Phase resonance $ \text{FRF} $	3.57	1
Double resonance RPM $ \text{FRF} $	0.328	1
Operating RPM $ \text{FRF} $	0.0217	1

B7. Normalized frequency response, 5000 samples

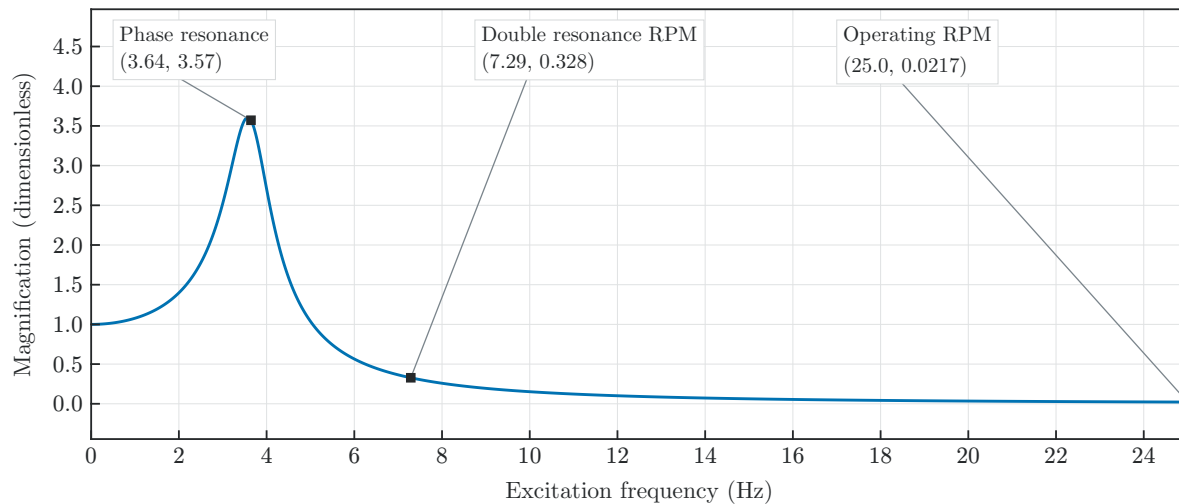


Figure 26: B.7. Dimensionless FRF magnitude and three requested frequency markers.

B7. Response phase relative to force

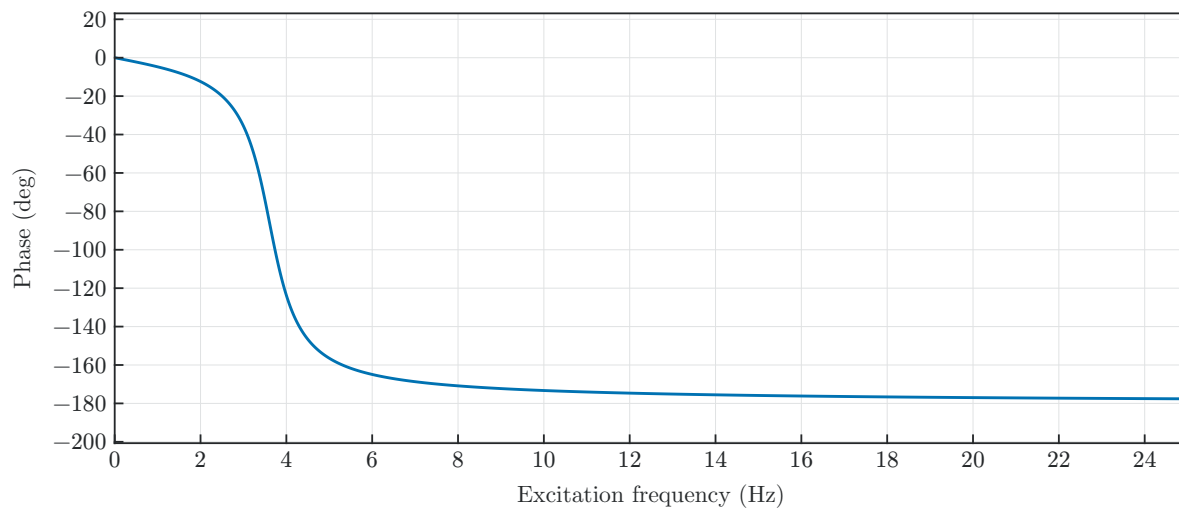


Figure 27: B.7. Phase relative to the harmonic force.

During a sufficiently slow run-up with the assumed constant force amplitude, the vibration grows near the response peak and falls as the operating speed moves above resonance. A rapid run-up need not reach the steady-state peak because the system takes time to build that response. An actual fixed rotating imbalance produces force proportional to speed squared, so a physical run-up prediction needs that speed dependence and an acceleration schedule. The homework supplies only one force amplitude; the plotted FRF is the structural response normalized by that force, not a complete run-up simulation.

B.8 Graduate Work. Calculate Horizontal Flex-Beam Response to Impact

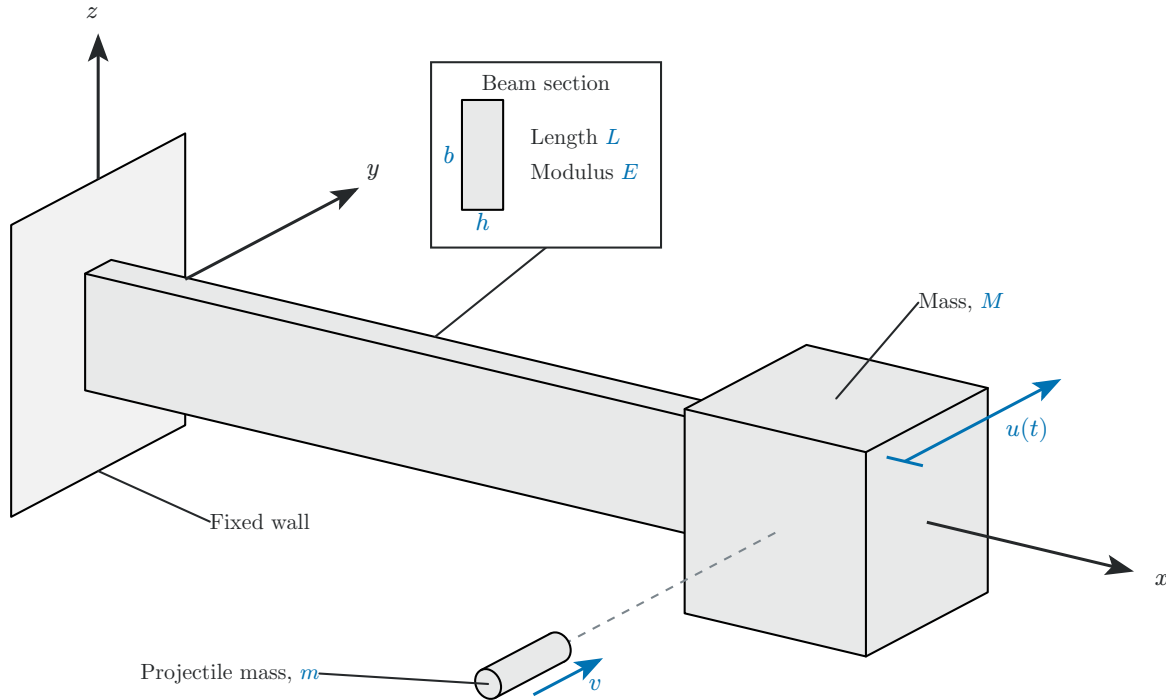


Figure 28: Impact excitation of a horizontal flex-beam vibration system.

Consider a cantilever beam of length $L = 568 \text{ mm}$ with a mass $M = 53 \text{ kg}$ attached at its tip as shown in Figure 28. The beam material is steel with $E = 200 \text{ GPa}$. The beam cross-section dimensions are $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. The beam is initially at rest. A small projectile of mass $m = 18 \text{ g}$ and velocity $v = 935 \text{ m/s}$ hits the mass and remains embedded in it. This impact imparts an initial velocity to the system that starts to oscillate with motion $u(t)$ contained in the horizontal plane.

Do the following.

- (1) Display input data.
- (2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$.
- (3) Calculate the cross-sectional stiffness EI of the beam.
- (4) Calculate the flex-beam spring constant k .
- (5) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass.
- (6) Calculate the rad/sec natural frequency ω_n and period of oscillation τ .
- (7) Calculate the position of the mass $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (8) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:
 - (a) The largest response.

- (b) The response calculated at item (7).
- (9) Discuss your results.

Worked solution

The inputs are $M = 53$ kg, $m_p = 0.018$ kg, $v_p = 935$ m/s, $L = 0.568$ m, $b = 0.037$ m, $h = 0.003$ m, and $E = 200$ GPa. The original sketch has section width h in the horizontal motion direction and section height b . Therefore the relevant moment of area is still

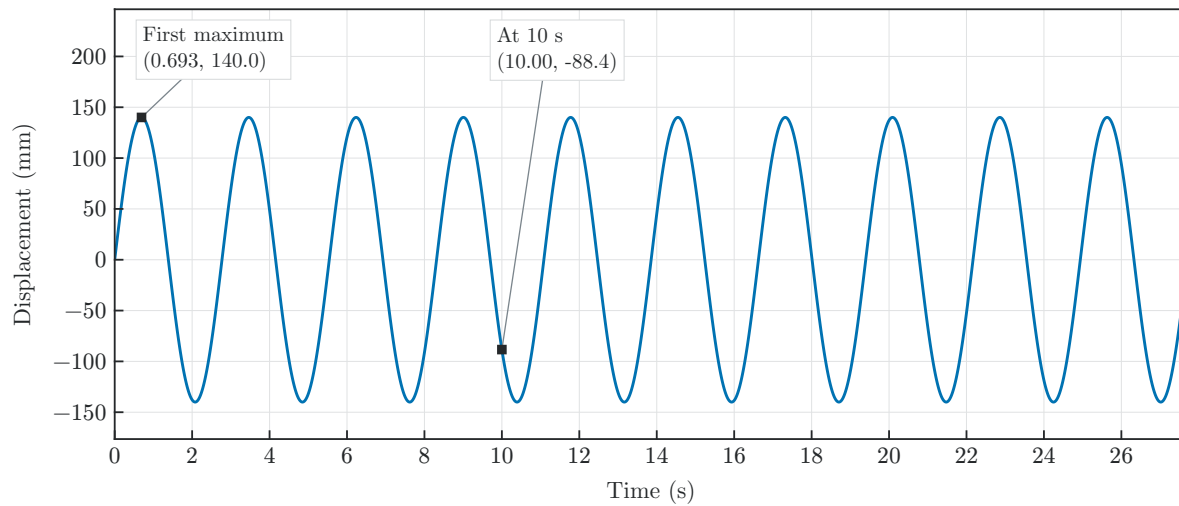
$$I = \frac{bh^3}{12}.$$

It is not $hb^3/12$ merely because the beam is moving horizontally. The cubed dimension is the section depth along the bending deflection. Under the short-impact lumped-mass approximation, the finite spring force has negligible impulse during the collision. Thus

$$v_0 = \frac{m_p v_p}{M + m_p}, \quad k = \frac{3EI}{L^3}, \quad \omega_n = \sqrt{\frac{k}{M + m_p}}.$$

With zero displacement immediately after impact, the response is again $u = (v_0/\omega_n) \sin(\omega_n t)$. The first largest positive response is at $\tau/4$.

Quantity	Value	Unit
I	83.2	mm ⁴
EI	16.65	N m ²
k	273	N/m
$M + m_p$	53.0	kg
v_0	0.317	m/s
ω_n	2.27	rad/s
f_n	0.361	Hz
τ	2.77	s
$u(10)$	-88.4	mm
$u_{\max}$	140.0	mm
$t_{\text{first peak}}$	0.693	s
10τ	27.7	s

B8. Response after an embedded-projectile impact**Figure 29:** *B.8. Weak-axis cantilever impact response.*

The predicted amplitude is a substantial fraction of the span. It should be reported as the requested linear-model result rather than an accurate large-deflection prediction. Beam inertia, local impact deformation, geometric nonlinearity, and damping are absent from this model.

3 Extra Credit Challenge Problems

Challenge problems are optional.

C.1 Analyze Vertical Spring-Mass Vibration System

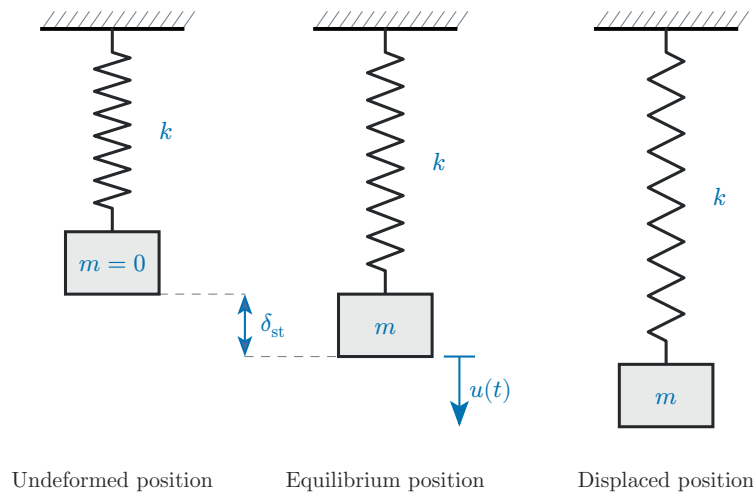


Figure 30: Vertical spring-mass vibration system.

Consider a vertical spring-mass vibration system consisting of a particle of mass m hanging on an elastic spring of stiffness k as shown in Figure 30.

Do the following.

- (1) Assume static equilibrium.
 - (a) Calculate the weight W acting on the mass m .
 - (b) Draw free-body diagram (FBD).
 - (c) Write equilibrium equation.
 - (d) Solve for the static displacement δ_{st} .
- (2) Assume vibration with time-varying vertical displacement $u(t)$.
 - (a) Draw free-body diagram (FBD).
 - (b) Deduce equation of motion (EOM).
 - (c) Compare the EOM deduced here with the EOM derived for horizontal spring-mass vibration system of Problem B.1.
- (3) Graduate work. Comment on how the EOM would change in the presence of an arbitrary force field acting at an angle θ with respect to the vertical axis.

Worked solution

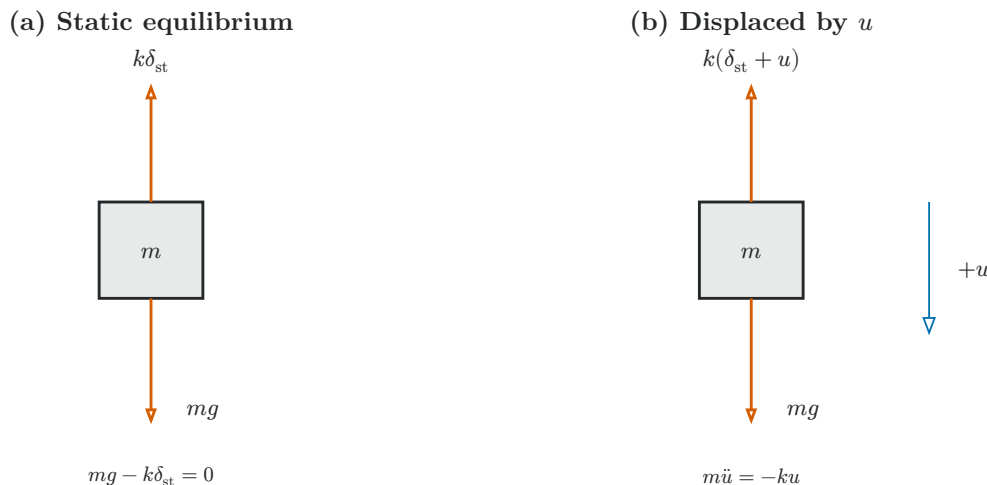


Figure 31: Free-body diagram for Problem C.1.

Let the downward coordinate y be measured from the spring's unstretched position. The weight is $W = mg$. At static equilibrium the upward spring force balances the weight,

$$mg - k\delta_{st} = 0, \quad \delta_{st} = \frac{mg}{k}.$$

During vibration, write $y = \delta_{st} + u$, where u is the displacement from static equilibrium. The upward spring force is $k(\delta_{st} + u)$, so

$$m\ddot{u} = mg - k(\delta_{st} + u) = (mg - k\delta_{st}) - ku = -ku.$$

Consequently $m\ddot{u} + ku = 0$, just as in B.1. Gravity still causes the static stretch; it disappears only from the incremental equation around that equilibrium.

For a force of magnitude F making angle θ with downward vertical, its component along the allowed coordinate is $F \cos \theta$. A constant uniform component only shifts equilibrium. If an added force component depends on time or position, it generally remains in the equation. With equilibrium y_e satisfying $ky_e = mg + F_y(y_e)$,

$$m\ddot{u} + ku = F_y(y_e + u, t) - F_y(y_e).$$

If the field is time independent and differentiable, linearization gives $m\ddot{u} + [k - F'_y(y_e)]u = 0$. The perpendicular component is carried by constraints only under the assumed one-coordinate model. This distinction is why an arbitrary force field cannot always be removed as a constant load.

C.2 Analyze Vertical Flex-Beam Vibration

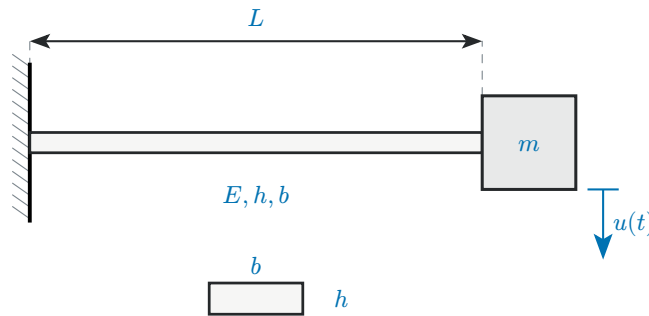


Figure 32: Vertical flex-beam vibrating system.

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 32. The beam cross-section properties are E , h , b .

Do the following.

- (1) Calculate the cross-sectional stiffness EI of the beam for vertical flexural deformation.
- (2) Calculate the spring constant k of the system.
- (3) Assume static equilibrium:
 - (a) Calculate the weight W acting on the mass m .
 - (b) Draw free-body diagram (FBD).
 - (c) Write equilibrium equation.
 - (d) Solve for the static displacement δ_{st} .
- (4) Assume vibration with time-varying vertical displacement $u(t)$ measured from the static equilibrium position:
 - (a) Draw free-body diagram (FBD).
 - (b) Deduce equation of motion (EOM).
 - (c) Compare the EOM deduced here with the EOM derived for horizontal flex-beam vibration system of Problem C.4.
- (5) Graduate work. Comment on how the EOM would change in the presence of an arbitrary force field acting at an angle θ with respect to the vertical axis.

Worked solution

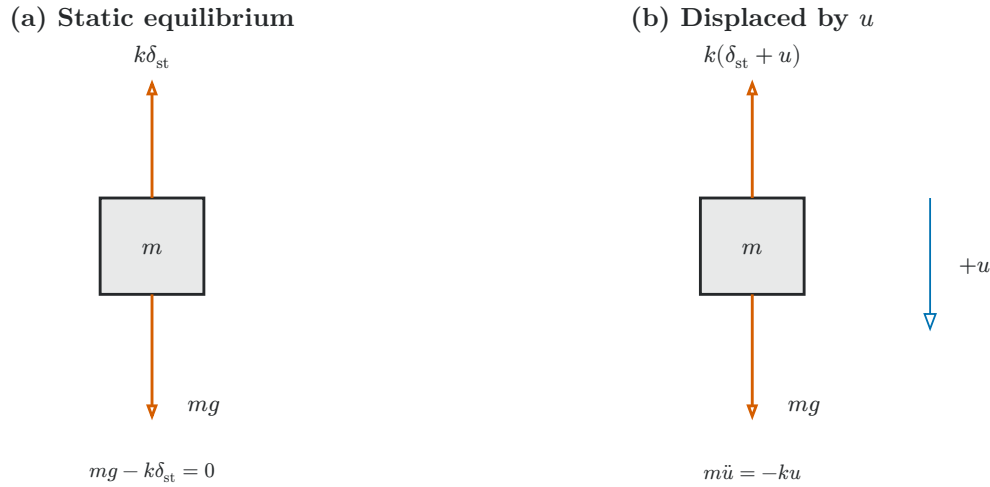


Figure 33: Free-body diagram for Problem C.2.

Use the rectangular section with vertical depth h and width b . Assuming small elastic bending of a uniform massless cantilever,

$$I = \frac{bh^3}{12}, \quad EI = E \frac{bh^3}{12}, \quad k = \frac{3EI}{L^3}.$$

At the tip, weight mg acts downward and the beam reaction $k\delta_{st}$ acts upward. Their balance gives $\delta_{st} = mg/k$. During vibration the upward beam force is $k(\delta_{st} + u)$. The same subtraction as in C.1 gives

$$m\ddot{u} + ku = 0, \quad \omega_n = \sqrt{\frac{3EI}{mL^3}}.$$

The normalized dynamic equation is identical to the horizontal-beam equation in C.3–C.4. The differences are the static offset and, when the section is rotated, which moment of area applies. The homework's C.4 cross-reference points to a numerical application; C.3 is the corresponding analytical derivation. For an arbitrary oblique field, project its component onto the bending direction and apply the equilibrium and linearization argument from C.1.

C.3 Analyze Horizontal Flex-Beam Vibration

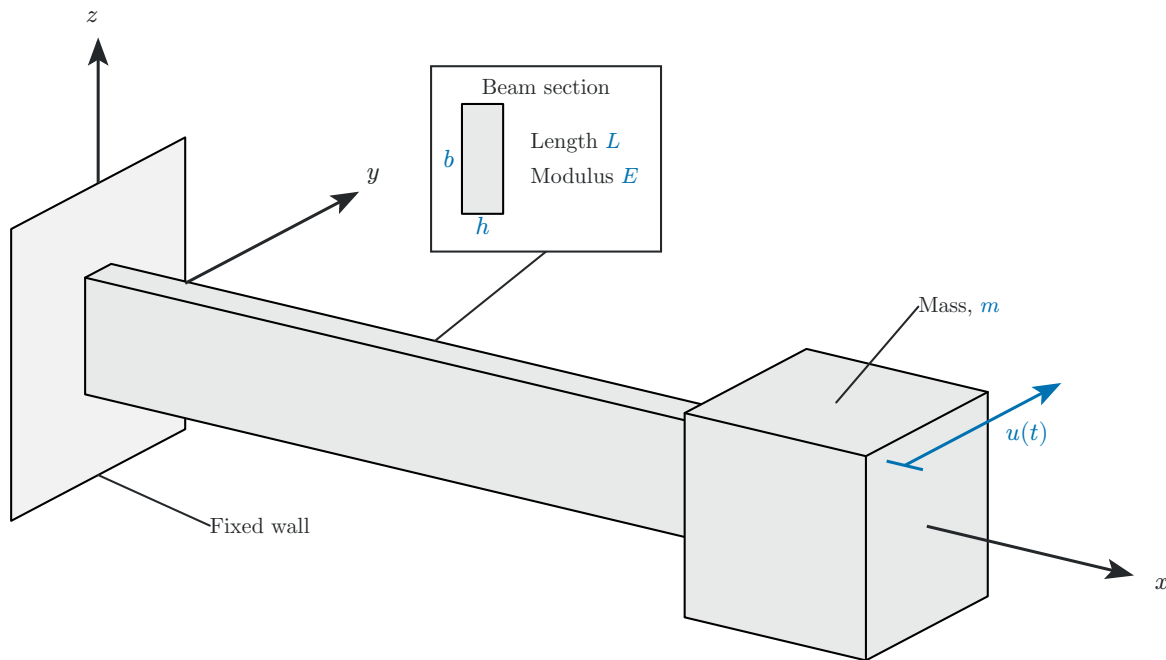


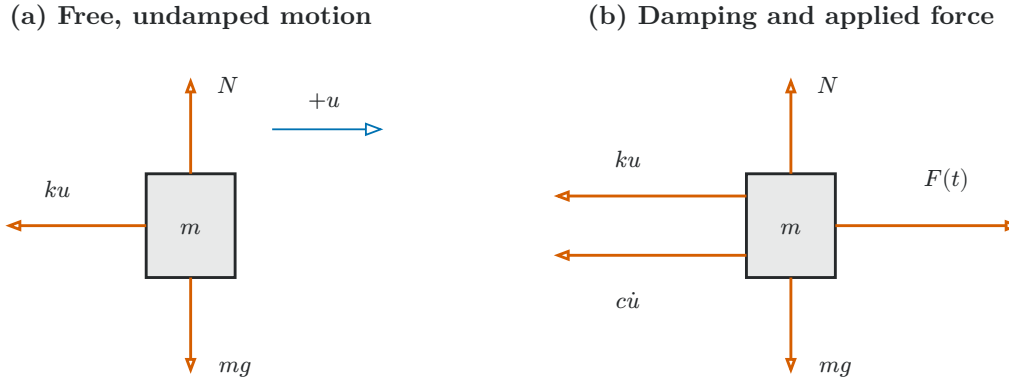
Figure 34: Horizontal flex-beam vibrating system.

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 34. The beam cross-section properties are E , h , b .

Do the following.

- (1) Assume vibration with horizontal time-varying displacement $u(t)$ as shown in Figure 34.
- (2) Calculate the cross-sectional stiffness EI of the beam for flexural deformation consistent with $u(t)$.
- (3) Calculate the flex-beam spring constant k .
- (4) Draw free-body diagram (FBD) including all the forces acting on the mass m .
- (5) Deduce equation of motion (EOM).
- (6) Compare the EOM deduced here with the EOM derived in Problem B.1.
- (7) Solve the initial value problem (IVP) and find the response for simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$.

Worked solution



Set $c = 0$ for an undamped system; set $F = 0$ for free vibration.

For beam models, N represents the balanced vertical support reaction.

Figure 35: Free-body diagram for Problem C.3.

In the original sketch, the section dimension along motion is h and the perpendicular dimension is b . Thus $I = bh^3/12$ and $k = 3EI/L^3$. On the free-body diagram, the horizontal elastic reaction is $-ku$. Weight is balanced by the beam's vertical support reaction about the static configuration, while the reduced horizontal dynamics satisfy

$$m\ddot{u} = -ku, \quad \ddot{u} + \frac{3EI}{mL^3}u = 0.$$

The equation has the same form as B.1. Define $\omega_n = \sqrt{3EI/(mL^3)}$ and impose the initial displacement and velocity to obtain

$$u(t) = u_0 \cos(\omega_n t) + \frac{v_0}{\omega_n} \sin(\omega_n t).$$

This assumes negligible coupling between horizontal motion and the static vertical deflection.

C.4 Calculate Horizontal Flex-Beam Response to Initial Displacement

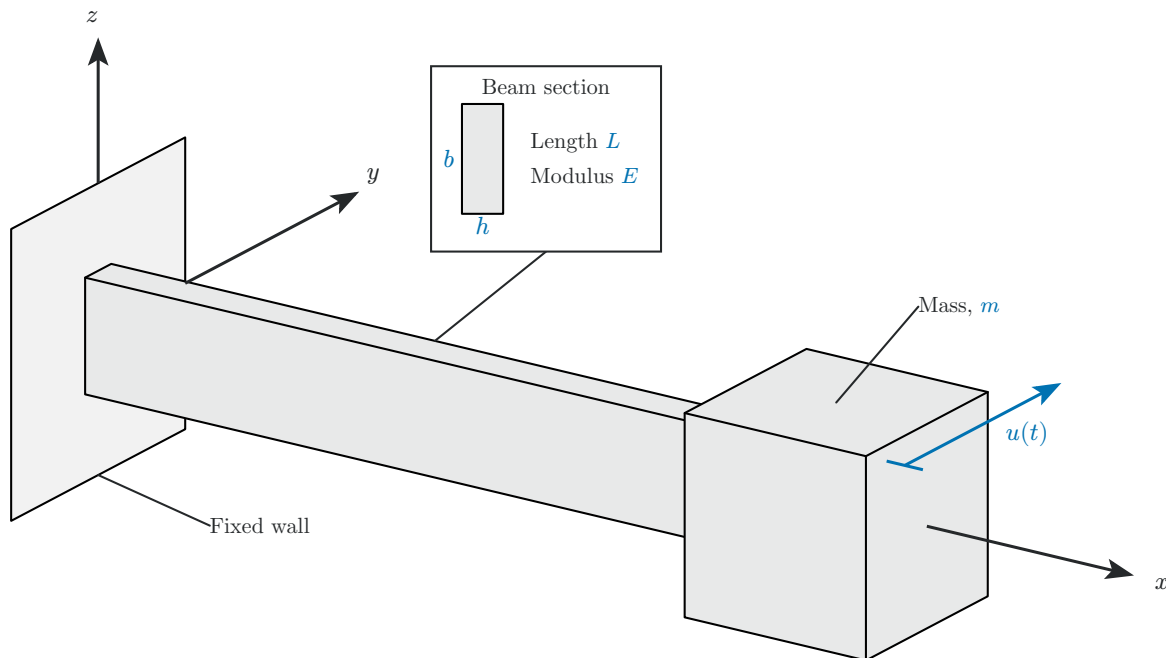


Figure 36: *Flex-beam vibrating system.*

Consider a cantilever beam of length $L = 568 \text{ mm}$ with a mass $m = 5.35 \text{ kg}$ attached at its tip as shown in Figure 36. The beam material is steel with $E = 200 \text{ GPa}$. The beam cross-section dimensions are $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. The tip mass is given an initial displacement $u_0 = 5.4 \text{ mm}$ with zero initial velocity.

Do the following.

- (1) Display input data.
- (2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$.
- (3) Calculate the cross-sectional stiffness EI of the beam.
- (4) Calculate the spring constant k of the system.
- (5) Calculate the natural frequency in rad/sec ω_n .
- (6) Calculate the natural frequency in Hz f_n .
- (7) Calculate the period of oscillation in sec τ .
- (8) Calculate the position of the mass $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (9) Plot the response of the system for $N_{\text{osc}} = 25$ oscillations and use a data tip to find $u(t_1)$ at t_1 .
- (10) Discuss your results.

Worked solution

The inputs are $m = 5.35$ kg, $L = 0.568$ m, $h = 0.003$ m, $b = 0.037$ m, $E = 200$ GPa, $u_0 = 5.4$ mm, and $v_0 = 0$. Apply C.3 and evaluate

$$u(t) = 0.0054 \cos(\omega_n t) \text{ m.}$$

Quantity	Value	Unit
I	83.2	mm ⁴
EI	16.65	N m ²
k	273	N/m
ω_n	7.14	rad/s
f_n	1.136	Hz
τ	0.880	s
$u(10)$	-3.45	mm
25τ	22.0	s

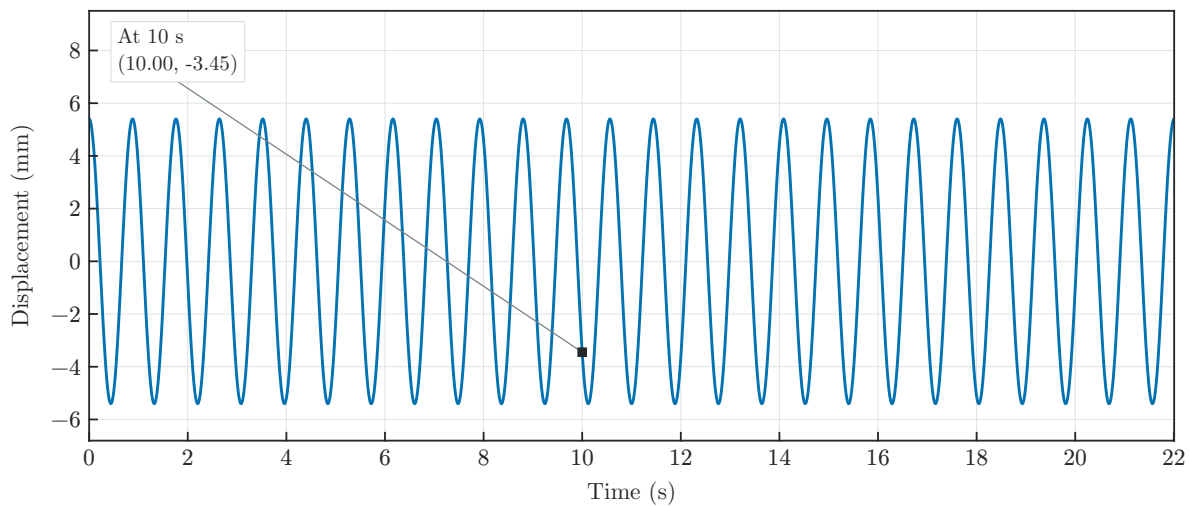
C4. Horizontal cantilever displacement release

Figure 37: *C.4. Twenty-five oscillations with the ten-second response marked.*

The maximum absolute displacement remains 5.4 mm because damping was not included. The response starts at a positive maximum, so a negative value at ten seconds is simply a later phase of the same oscillation.

C.5 Analyze Spring-Mass Forced Vibration

Consider a spring-mass vibrating system excited by a time-dependent force $F(t)$ as shown in Figure 38.

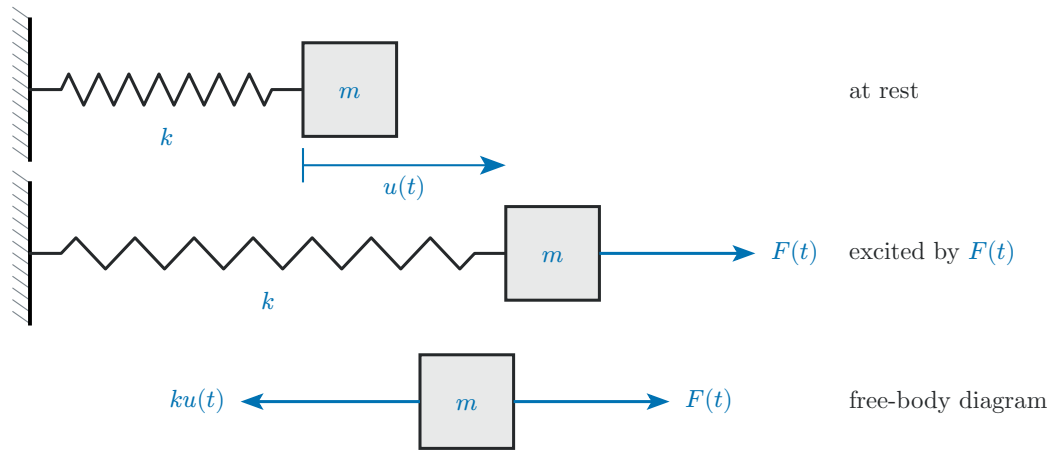


Figure 38: Spring-mass system excited by force $F(t)$.

General Analysis of Spring-Mass Forced Vibration

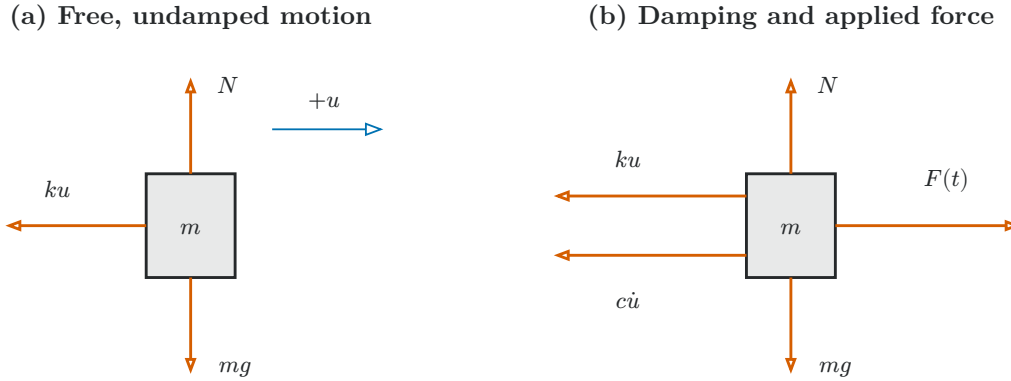
Do the following.

- (1) State your assumptions in performing this analysis.
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.
- (3) Write NLM equation.
- (4) Deduce the equation of motion (EOM) and express it as a nonhomogeneous ODE in terms of m , k , $F(t)$.
- (5) Write the ODE general solution as the superposition of a complementary solution $u_c(t)$ and a particular solution $u_p(t)$.
- (6) Recall the complementary solution $u_c(t)$ as the solution of the homogeneous equation derived in Problem B.1 and write it in terms of trigonometric functions multiplied by constants A and B .

Spring-Mass Response to a Suddenly Applied Load

- (7) Assume the loading $F(t)$ is a suddenly applied load of magnitude F_0 .
- (8) Find the particular solution for this case of a suddenly applied load.
- (9) Write the complete solution as the sum of $u_c(t)$ from (6) and $u_p(t)$ from (8).
- (10) Assume “at rest” initial conditions to find the constants A and B and write the complete solution $u(t)$ as a function of F_0 , k , and $\cos \omega_n t$.
- (11) Use (10) to find the largest displacement $u_{\max}$ and compare it with the static displacement u_{st} .

Worked solution



Set $c = 0$ for an undamped system; set $F = 0$ for free vibration.

For beam models, N represents the balanced vertical support reaction.

Figure 39: Free-body diagram for Problem C.5.

Assume the B.1 model with a prescribed force $F(t)$. The free-body diagram gives $m\ddot{u} = F(t) - ku$, so

$$m\ddot{u} + ku = F(t).$$

The general response is $u = u_c + u_p$, with $u_c = A \cos(\omega_n t) + B \sin(\omega_n t)$. For a constant load F_0 switched on at time zero and held thereafter, a constant particular solution satisfies $ku_p = F_0$, so $u_p = F_0/k$.

Before release the mass is at rest at the unloaded position. Thus $u(0) = 0$ and $\dot{u}(0) = 0$, giving $A = -F_0/k$ and $B = 0$. The complete response for nonnegative time is

$$u(t) = \frac{F_0}{k} [1 - \cos(\omega_n t)].$$

The static displacement is $u_{st} = F_0/k$. The first maximum occurs when $\cos(\omega_n t) = -1$, at $t = \pi/\omega_n$, and equals $u_{max} = 2F_0/k = 2u_{st}$.

Physically, the mass passes through static equilibrium with nonzero velocity. It continues until the spring has stored all the work done by the load. The turning-point energy balance $\frac{1}{2}ku_{max}^2 = F_0u_{max}$ independently yields the same nonzero root. A suddenly applied load is a step input, not an impulse.

C.6 Calculate Cantilever Response to Suddenly Applied Weight

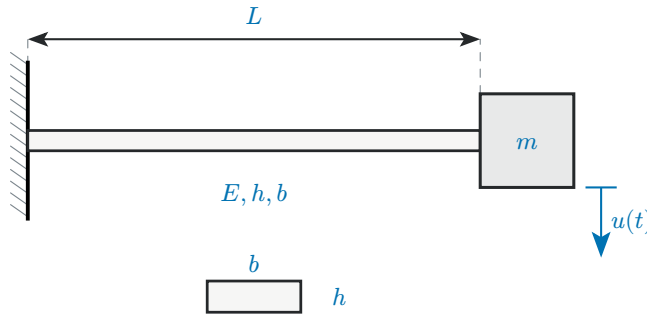


Figure 40: Cantilever beam with tip mass.

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a cantilever beam of length $L = 568 \text{ mm}$ and a mass $m = 5.35 \text{ kg}$ that has to be attached to its tip as shown in Figure 40. The beam is made of steel with elastic modulus $E = 200 \text{ GPa}$, yield stress $Y = 710 \text{ MPa}$, and cross-section dimensions $h = 3 \text{ mm}$ and $b = 37 \text{ mm}$.

Do the following.

- (1) Display input data.
- (2) Calculate the cross-section second moment of area I , the flexural stiffness EI , the spring constant k , the natural frequency ω_n rad/s, f_n Hz, and period of oscillation τ of the system.
- (3) Calculate the weight F_0 of the mass m .
- (4) Static case. Assume the mass is attached to the undeformed beam and then released gradually without inducing vibration. Calculate the static displacement u_{st} .
- (5) Dynamic case. Assume the mass is attached to the undeformed beam and then released suddenly, inducing vibration.
 - (a) Calculate the maximum dynamic displacement u_{dyn} .
 - (b) Plot the response $u(t)$ measured from the undeformed position for $N_{\text{osc}} = 10$ oscillations.
 - (c) Place a data tip on the plot to identify the maximum displacement and compare it with the result of item (5)(a).
- (6) Graduate work. Calculate the maximum stress at the root of the beam and the safety factor $SF = Y/\sigma$, and comment on your results for the following two cases.
 - (a) Static-case stress σ_{st} .
 - (b) Dynamic-case stress σ_{dyn} .
- (7) Discuss your results.

Worked solution

The inputs are $L = 0.568$ m, $m = 5.35$ kg, $E = 200$ GPa, $h = 0.003$ m, $b = 0.037$ m, $g = 9.81$ m/s², and yield stress $Y = 710$ MPa. The mass is already attached at rest to the undeflected tip before release. There is no drop height or impact velocity in this model.

Use $I = bh^3/12$, $k = 3EI/L^3$, $\omega_n = \sqrt{k/m}$, and $F_0 = mg$. A gradual release gives $u_{\text{st}} = mg/k$. A sudden release gives $u(t) = (mg/k)[1 - \cos(\omega_n t)]$ from the unloaded position.

For the graduate stress calculation, the static root moment is $M_{\text{root}} = F_0 L$. The outer-fibre distance is $h/2$, giving

$$\sigma_{\text{st}} = \frac{F_0 L (h/2)}{I}.$$

At the dynamic maximum, the elastic restoring tip force is $ku_{\text{max}} = 2F_0$. For the assumed massless elastic beam, the root moment and bending stress are therefore doubled,

$$\sigma_{\text{dyn}} = 2\sigma_{\text{st}}, \quad \text{SF}_{\text{st}} = \frac{Y}{\sigma_{\text{st}}}, \quad \text{SF}_{\text{dyn}} = \frac{Y}{2\sigma_{\text{st}}}.$$

Quantity	Value	Unit
I	83.2	mm ⁴
EI	16.65	N m ²
k	273	N/m
ω_n	7.14	rad/s
f_n	1.136	Hz
τ	0.880	s
F_0	52.5	N
u_{st}	192.5	mm
$u_{\text{dyn,max}}$	385	mm
σ_{st}	537	MPa
σ_{dyn}	1074	MPa
SF_{st}	1.322	1
SF_{dyn}	0.661	1
$u_{\text{dyn,max}}/L$	0.678	1

C6. Ideal linear response to suddenly applied weight

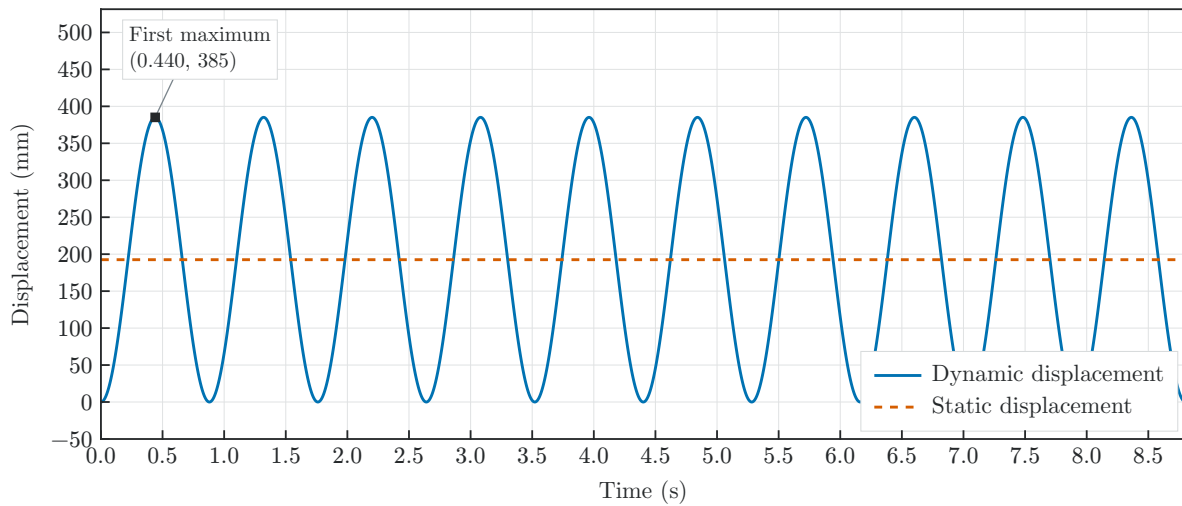


Figure 41: C.6. Formal linear response, with the first twice-static maximum marked.

The dynamic safety factor is below one, so the linear elastic stress calculation predicts that yielding would occur. In addition, the predicted deflection is large relative to the beam length. Once yielding or geometric nonlinearity matters, the elastic stiffness and sinusoidal solution are no longer quantitatively reliable. The appropriate conclusion is that the given ideal model flags a problem, not that the plotted large-amplitude motion is an accurate prediction of the real beam.

C.7 Analyze Flex-Beam Pendulum

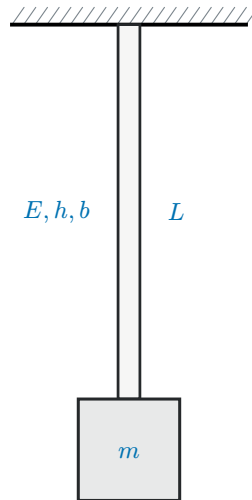


Figure 42: *Flex-beam pendulum.*

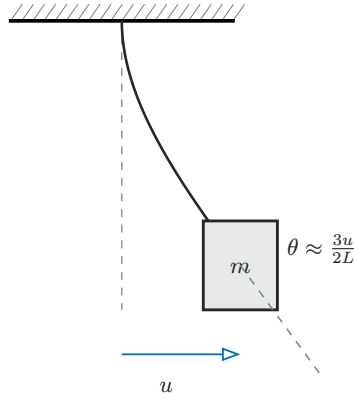
Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 42. The beam is fixed into the ceiling and is hanging downward along the z axis. The mass may oscillate sideways in either x or y directions. The beam cross-section properties are E, h, b .

Do the following.

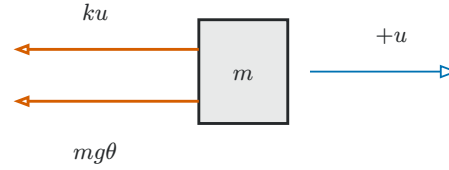
- (1) Calculate the minor and major second moments of area I_1 and I_2 .
- (2) Assume a horizontal displacement $u(t)$ which produces flexure about the minor I_1 .
- (3) Draw free-body diagram (FBD).
- (4) Write Newton's law of motion (NLM) equation.
- (5) Apply small-angle approximation and deduce the equation of motion (EOM). Clearly state your assumptions.
- (6) Define the natural frequency $\omega_{n,\text{FBP}}$ of the flex-beam pendulum.
- (7) Deduce the governing equation.
- (8) Compare your results with those of Problem B.1 and write the general solution.
- (9) Solve the initial value problem (IVP), i.e., deduce the response for simultaneous application of initial displacement u_0 and initial velocity $\dot{u}_0$.
- (10) Graduate work. Explain how the solution would change if flexure were about I_2 .

Worked solution

(a) Bending shape and tip slope



(b) Reduced restoring projection



Course-note approximation, PDF page 252.

$$m\ddot{u} + \left(k + \frac{3mg}{2L}\right)u = 0$$

Figure 43: Free-body diagram for Problem C.7.

For a rectangular section with $b > h$, the minor and major second moments of area are

$$I_1 = \frac{bh^3}{12}, \quad I_2 = \frac{hb^3}{12}.$$

Use I_1 for weak-axis motion. Page 252 combines $k = 3EI_1/L^3$ with gravity using the beam-tip slope θ . The assumed static tip-load bending shape gives

$$u = \frac{PL^3}{3EI_1}, \quad \theta = \frac{PL^2}{2EI_1}, \quad \frac{\theta}{u} = \frac{3}{2L}.$$

The note's projected force balance is $-ku - mg\theta = m\ddot{u}$. With $\theta \approx 3u/(2L)$, it gives

$$m\ddot{u} + \left(k + \frac{3mg}{2L}\right)u = 0, \quad \omega_{n,\text{FBP}} = \sqrt{\frac{3EI_1}{mL^3} + \frac{3g}{2L}}.$$

The governing equation is $\ddot{u} + \omega_{n,\text{FBP}}^2 u = 0$, so its general solution is $A \cos(\omega_{n,\text{FBP}}t) + B \sin(\omega_{n,\text{FBP}}t)$. The initial-value response is

$$u(t) = u_0 \cos(\omega_{n,\text{FBP}}t) + \frac{v_0}{\omega_{n,\text{FBP}}} \sin(\omega_{n,\text{FBP}}t).$$

This approximation assumes a fixed bending shape, small slope, negligible beam mass, and no damping. The $3g/(2L)$ term uses the tip-slope projection on page 252, not the ordinary pendulum angle u/L . A continuum or energy-consistent tensioned-beam model can give a different gravitational stiffness.

For major-axis bending, substitute I_2 for I_1 . Elastic stiffness increases by $(b/h)^2$. The frequency increases, approaching a ratio of b/h only when elastic stiffness dominates gravity in both cases.

C.8 Calculate Flex-Beam Pendulum Response to Impact

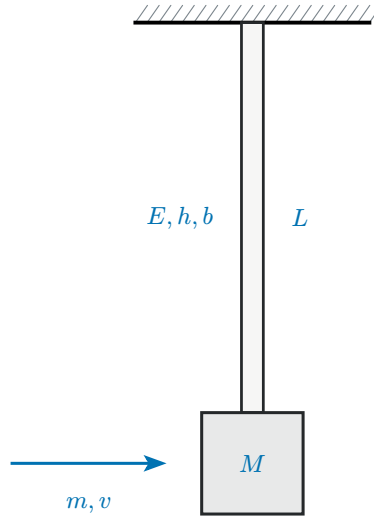


Figure 44: Impact excitation of a vertical flex-beam pendulum.

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a vertical flex-beam pendulum of length $L = 568 \text{ mm}$ with a mass $M = 42 \text{ kg}$ attached at its end, as shown in Figure 44. The beam is steel with $E = 200 \text{ GPa}$ and cross-section dimensions $h = 3 \text{ mm}$ and $b = 37 \text{ mm}$. The beam is initially at rest. A projectile of mass $m = 18 \text{ g}$ and velocity $v = 935 \text{ m/s}$ hits the mass and remains embedded in it. The projectile orientation causes the beam to flex about its minor cross-sectional axis. The impact imparts an initial velocity to the system, which then oscillates.

Do the following.

- (1) Display input data.
- (2) Calculate the minor second moment of area I of the beam.
- (3) Calculate the minor cross-sectional stiffness EI .
- (4) Calculate the flex-beam spring constant k .
- (5) Calculate the initial velocity of the system $\dot{u}_0$ as resulting from the interaction between the projectile and the large mass.
- (6) Calculate the rad/s natural frequency $\omega_{n,\text{FBP}}$ and period of oscillation τ_{FBP} of the flex-beam pendulum with projectile embedded into it.
- (7) Calculate the response $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (8) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:
 - (a) The largest response.
 - (b) The response value calculated at item (7).
- (9) Compare your results with those of Problem A.3.

Worked solution

The inputs are $L = 0.568$ m, $M = 42$ kg, $m_p = 0.018$ kg, $v_p = 935$ m/s, $E = 200$ GPa, $h = 0.003$ m, $b = 0.037$ m, and $g = 9.81$ m/s². The specified projectile orientation excites the minor axis.

Use the same short-duration collision assumption as B.8. Calculate the post-impact velocity from $v_0 = m_p v_p / (M + m_p)$, then insert the total mass into the C.7 frequency,

$$\omega_{n,\text{FBP}} = \sqrt{\frac{k}{M + m_p} + \frac{3g}{2L}}, \quad u(t) = \frac{v_0}{\omega_{n,\text{FBP}}} \sin(\omega_{n,\text{FBP}} t).$$

Quantity	Value	Unit
I	83.2	mm ⁴
EI	16.65	N m ²
k	273	N/m
$M + m_p$	42.0	kg
v_0	0.401	m/s
ω_n	5.69	rad/s
f_n	0.906	Hz
τ	1.104	s
$u(10)$	25.2	mm
u_{max}	70.4	mm
$t_{\text{first peak}}$	0.276	s
10τ	11.04	s

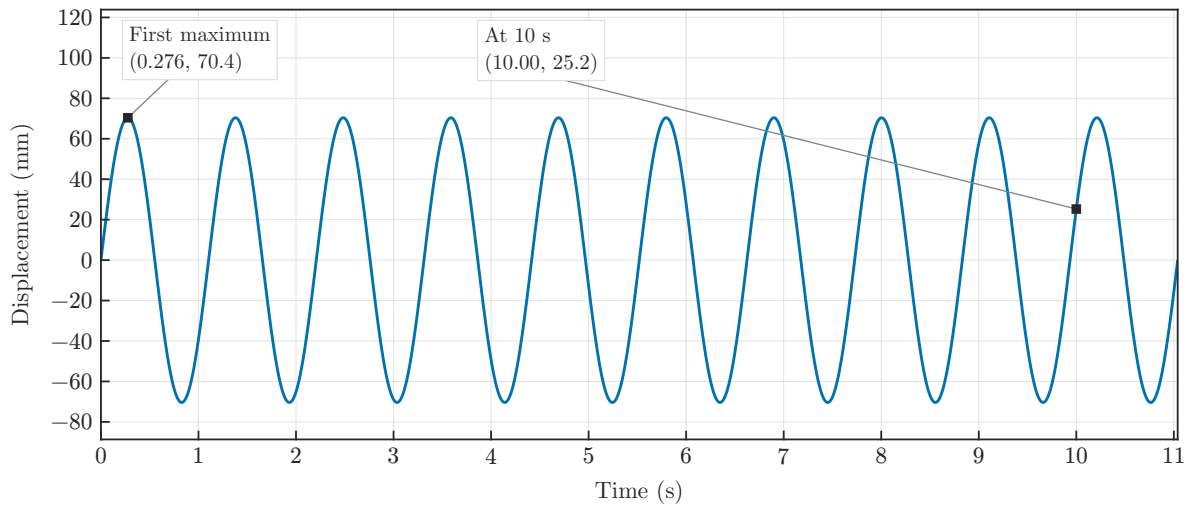
C8. Response after an embedded-projectile impact

Figure 45: C.8. Response under the approximation on note page 252.

Compared with A.3, the post-impact speed is similar while the frequency is higher, giving a smaller amplitude and shorter period. The two problems also differ in mass, length, projectile data, and gravity approximation, so this comparison does not isolate beam stiffness. The ten-second displacements depend on phase as well as amplitude.

C.9 Complex Exponentials and Trigonometric Functions

- (1) Find the solutions of $z^4 = 1$ (the fourth roots of unity) and express them as:
 - (a) Complex numbers.
 - (b) Complex exponentials.
- (2) Write the complex number $e^{i\omega t}$ as a sum of trigonometric functions (hint, Euler's formula).
- (3) Calculate the phase difference ϕ in radians and degrees between the following signals:
 - (a) Compare $x_1(t) = \sin(\omega t)$ and $x_2(t) = \sin(\omega t + 3\pi/4)$.
 - (b) Compare $x_1(t) = \sin(\omega t + \pi/3)$ and $x_2(t) = \sin(\omega t + 3\pi/4)$.
 - (c) Compare $x_1(t) = \sin(\omega t + 45^\circ)$ and $x_2(t) = \sin(\omega t + 115^\circ)$.
- (4) What is the phase in radians and degrees of the following complex numbers:
 - (a) $e^{i\pi/2}$.
 - (b) $e^{i3\pi/2}$.
 - (c) $e^{i\pi/4}$.
 - (d) $e^{-i\pi/2}$.
 - (e) $e^{-i\pi/4}$.
- (5) Draw the locations on the unit circle of the complex numbers listed in item (4).
- (6) Calculate the magnitude and phase in degrees of the complex number $Z = 4 + 5i$.

Worked solution

Write a complex number of unit magnitude as $z = e^{i\theta}$. Then $z^4 = 1$ requires $4\theta = 2\pi n$. The four distinct roots are

$$z_n = e^{in\pi/2}, \quad n = 0, 1, 2, 3,$$

which are 1 , i , -1 , and $-i$. Euler's formula is

$$e^{i\omega t} = \cos(\omega t) + i \sin(\omega t).$$

Define the requested signal phase difference as the phase of the second signal minus the phase of the first. A positive value means that the second sinusoid leads the first. For the three signal pairs, the answers are respectively

$$\Delta\phi_a = \frac{3\pi}{4} = 135^\circ,$$

$$\Delta\phi_b = \frac{3\pi}{4} - \frac{\pi}{3} = \frac{5\pi}{12} = 75^\circ,$$

$$\Delta\phi_c = 115^\circ - 45^\circ = 70^\circ = \frac{7\pi}{18}.$$

The phases of $e^{i\pi/2}$, $e^{i3\pi/2}$, $e^{i\pi/4}$, $e^{-i\pi/2}$, and $e^{-i\pi/4}$ are $\pi/2$, $3\pi/2$, $\pi/4$, $-\pi/2$, and $-\pi/4$, respectively, or 90° , 270° , 45° , -90° , and -45° . The principal phase of $e^{i3\pi/2}$ is -90° , and that point coincides with $e^{-i\pi/2}$. Angles differing by a full revolution represent the same complex number.

Finally, for $Z = 4 + 5i$,

$$|Z| = \sqrt{4^2 + 5^2} = \sqrt{41} \approx 6.40, \quad \arg Z = \text{atan2}(5, 4) \approx 51.3^\circ.$$

Complex phases on the unit circle

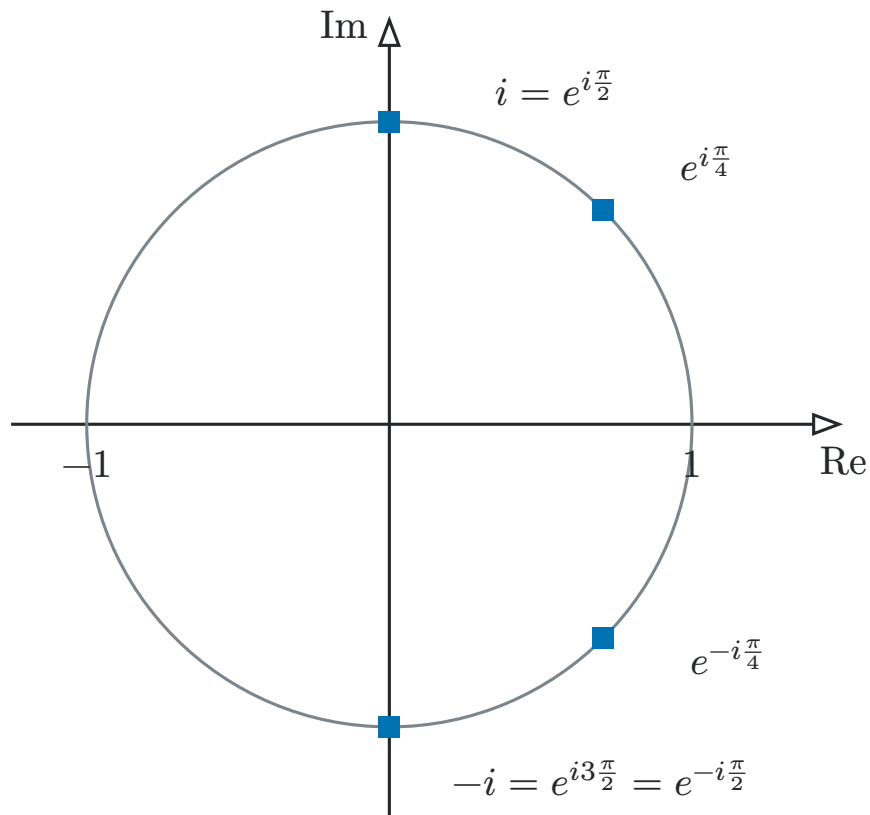


Figure 46: C.9. The requested complex numbers on the unit circle.